

Engineering Materials

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Classification of engineering materials

Due to dominant bonds:

- Metals and alloys
- Ceramics
- Polymers
- Composites

General properties of metals

- High electrical conductivity decreasing with increase of temperature,
- Good thermal conductivity,
- Ductility (toughness),
- Metallic sheen,
- No transparency for light.

Why metals are important?

- They have properties that satisfy a wide variety of design requirements
- The manufacturing processes by which they are shaped into products have been developed and refined over many years
- Engineers understand metals
- High stiffness and strength - can be alloyed for high rigidity, strength, and hardness
- Toughness - capacity to absorb energy better than other classes of materials
- Cost - the price of steel is very competitive with other engineering materials

Major metals

Metal	Density Mg/m ³	Melting temperature °C
Fe	7.87	1538
Al	2.70	660
Cu	8.96	1084
Ti	4.50	1668

Production of metals

Metal	Annual production mln tons	Energy consumption GJ/ton
Fe (steel)	1 200	50 (from metal ore) 17 (from scrap-metal) 35 (oxygen convector furnance)
Al	63 20	300 (from bauxite) 15 (recycling)
Cu	21	90 (pirometallurgy)
Ti	0.22	900 (from ore)
polymers	300	100

Why metals are important?

- Some metals are important as pure elements (gold, silver, copper)
- Other pure metals are not used very often. Titanium is used in a pure form where corrosion resistance is the most important factor
- Most engineering applications require the enhanced properties obtained by alloying
- Through alloying, it is possible to increase strength, hardness, and other properties compared to pure metals

Metals

Periodic Table of the Elements

1	I A	1	H	II A	2	0	He
2		3	Li	4	Be		
3		11	Na	12	Mg		
4		19	K	20	Ca	II B	
5		37	Rb	38	Sr	Y	
6		55	Cs	56	Ba	* La	
7		87	Fr	88	Ra	+ Ac	

The diagram illustrates the periodic table with the following structure:

- Metals (Green):** Elements 1 through 12 are in the first row. Elements 13 through 18 are in the second row. Elements 19 through 36 are in the third row. Elements 37 through 54 are in the fourth row. Elements 55 through 86 are in the fifth row. Elements 87 through 118 are in the sixth row. Elements 119 through 150 are in the seventh row. Elements 151 through 180 are in the eighth row. Elements 181 through 210 are in the ninth row. Elements 211 through 240 are in the tenth row. Elements 241 through 270 are in the eleventh row. Elements 271 through 300 are in the twelfth row. Elements 301 through 330 are in the thirteenth row. Elements 331 through 360 are in the fourteenth row. Elements 361 through 390 are in the fifteenth row. Elements 391 through 420 are in the sixteenth row. Elements 421 through 450 are in the seventeenth row. Elements 451 through 480 are in the eighteenth row. Elements 481 through 510 are in the nineteenth row. Elements 511 through 540 are in the twentieth row. Elements 541 through 570 are in the twenty-first row. Elements 571 through 600 are in the twenty-second row. Elements 601 through 630 are in the twenty-third row. Elements 631 through 660 are in the twenty-fourth row. Elements 661 through 690 are in the twenty-fifth row. Elements 691 through 720 are in the twenty-sixth row. Elements 721 through 750 are in the twenty-seventh row. Elements 751 through 780 are in the twenty-eighth row. Elements 781 through 810 are in the twenty-ninth row. Elements 811 through 840 are in the thirtieth row. Elements 841 through 870 are in the thirty-first row. Elements 871 through 900 are in the thirty-second row. Elements 901 through 930 are in the thirty-third row. Elements 931 through 960 are in the thirty-fourth row. Elements 961 through 990 are in the thirty-fifth row. Elements 991 through 1020 are in the thirty-sixth row. Elements 1021 through 1050 are in the thirty-seventh row. Elements 1051 through 1080 are in the thirty-eighth row. Elements 1081 through 1110 are in the thirty-ninth row. Elements 1111 through 1140 are in the fortieth row. Elements 1141 through 1170 are in the forty-first row. Elements 1171 through 1200 are in the forty-second row. Elements 1201 through 1230 are in the forty-third row. Elements 1231 through 1260 are in the forty-fourth row. Elements 1261 through 1290 are in the forty-fifth row. Elements 1291 through 1320 are in the forty-sixth row. Elements 1321 through 1350 are in the forty-seventh row. Elements 1351 through 1380 are in the forty-eighth row. Elements 1381 through 1410 are in the forty-ninth row. Elements 1411 through 1440 are in the fiftieth row. Elements 1441 through 1470 are in the fifty-first row. Elements 1471 through 1500 are in the fifty-second row. Elements 1501 through 1530 are in the fifty-third row. Elements 1531 through 1560 are in the fifty-fourth row. Elements 1561 through 1590 are in the fifty-fifth row. Elements 1591 through 1620 are in the fifty-sixth row. Elements 1621 through 1650 are in the fifty-seventh row. Elements 1651 through 1680 are in the fifty-eighth row. Elements 1681 through 1710 are in the fifty-ninth row. Elements 1711 through 1740 are in the sixtieth row. Elements 1741 through 1770 are in the sixty-first row. Elements 1771 through 1800 are in the sixty-second row. Elements 1801 through 1830 are in the sixty-third row. Elements 1831 through 1860 are in the sixty-fourth row. Elements 1861 through 1890 are in the sixty-fifth row. Elements 1891 through 1920 are in the sixty-sixth row. Elements 1921 through 1950 are in the sixty-seventh row. Elements 1951 through 1980 are in the sixty-eighth row. Elements 1981 through 2010 are in the sixty-ninth row. Elements 2011 through 2040 are in the seventieth row. Elements 2041 through 2070 are in the seventy-first row. Elements 2071 through 2100 are in the seventy-second row. Elements 2101 through 2130 are in the seventy-third row. Elements 2131 through 2160 are in the seventy-fourth row. Elements 2161 through 2190 are in the seventy-fifth row. Elements 2191 through 2220 are in the seventy-sixth row. Elements 2221 through 2250 are in the seventy-seventh row. Elements 2251 through 2280 are in the seventy-eighth row. Elements 2281 through 2310 are in the seventy-ninth row. Elements 2311 through 2340 are in the eightieth row. Elements 2341 through 2370 are in the eighty-first row. Elements 2371 through 2400 are in the eighty-second row. Elements 2401 through 2430 are in the eighty-third row. Elements 2431 through 2460 are in the eighty-fourth row. Elements 2461 through 2490 are in the eighty-fifth row. Elements 2491 through 2520 are in the eighty-sixth row. Elements 2521 through 2550 are in the eighty-seventh row. Elements 2551 through 2580 are in the eighty-eighth row. Elements 2581 through 2610 are in the eighty-ninth row. Elements 2611 through 2640 are in the ninetieth row. Elements 2641 through 2670 are in the ninety-first row. Elements 2671 through 2700 are in the ninety-second row. Elements 2701 through 2730 are in the ninety-third row. Elements 2731 through 2760 are in the ninety-fourth row. Elements 2761 through 2790 are in the ninety-fifth row. Elements 2791 through 2820 are in the ninety-sixth row. Elements 2821 through 2850 are in the ninety-seventh row. Elements 2851 through 2880 are in the ninety-eighth row. Elements 2881 through 2910 are in the ninety-ninth row. Elements 2911 through 2940 are in the one hundredth row. Elements 2941 through 2970 are in the one hundred first row. Elements 2971 through 3000 are in the one hundred second row. Elements 3001 through 3030 are in the one hundred third row. Elements 3031 through 3060 are in the one hundred fourth row. Elements 3061 through 3090 are in the one hundred fifth row. Elements 3091 through 3120 are in the one hundred sixth row. Elements 3121 through 3150 are in the one hundred seventh row. Elements 3151 through 3180 are in the one hundred eighth row. Elements 3181 through 3210 are in the one hundred ninth row. Elements 3211 through 3240 are in the one hundred tenth row. Elements 3241 through 3270 are in the one hundred eleventh row. Elements 3271 through 3300 are in the one hundred twelfth row. Elements 3301 through 3330 are in the one hundred thirteenth row. Elements 3331 through 3360 are in the one hundred fourteenth row. Elements 3361 through 3390 are in the one hundred fifteenth row. Elements 3391 through 3420 are in the one hundred sixteenth row. Elements 3421 through 3450 are in the one hundred seventeenth row. Elements 3451 through 3480 are in the one hundred eighteenth row. Elements 3481 through 3510 are in the one hundred nineteenth row. Elements 3511 through 3540 are in the one hundred twentieth row. Elements 3541 through 3570 are in the one hundred twenty-first row. Elements 3571 through 3600 are in the one hundred twenty-second row. Elements 3601 through 3630 are in the one hundred twenty-third row. Elements 3631 through 3660 are in the one hundred twenty-fourth row. Elements 3661 through 3690 are in the one hundred twenty-fifth row. Elements 3691 through 3720 are in the one hundred twenty-sixth row. Elements 3721 through 3750 are in the one hundred twenty-seventh row. Elements 3751 through 3780 are in the one hundred twenty-eighth row. Elements 3781 through 3810 are in the one hundred twenty-ninth row. Elements 3811 through 3840 are in the one hundred thirtieth row. Elements 3841 through 3870 are in the one hundred thirty-first row. Elements 3871 through 3900 are in the one hundred thirty-second row. Elements 3901 through 3930 are in the one hundred thirty-third row. Elements 3931 through 3960 are in the one hundred thirty-fourth row. Elements 3961 through 3990 are in the one hundred thirty-fifth row. Elements 3991 through 4020 are in the one hundred thirty-sixth row. Elements 4021 through 4050 are in the one hundred thirty-seventh row. Elements 4051 through 4080 are in the one hundred thirty-eighth row. Elements 4081 through 4110 are in the one hundred thirty-ninth row. Elements 4111 through 4140 are in the one hundred fortieth row. Elements 4141 through 4170 are in the one hundred forty-first row. Elements 4171 through 4200 are in the one hundred forty-second row. Elements 4201 through 4230 are in the one hundred forty-third row. Elements 4231 through 4260 are in the one hundred forty-fourth row. Elements 4261 through 4290 are in the one hundred forty-fifth row. Elements 4291 through 4320 are in the one hundred forty-sixth row. Elements 4321 through 4350 are in the one hundred forty-seventh row. Elements 4351 through 4380 are in the one hundred forty-eighth row. Elements 4381 through 4410 are in the one hundred forty-ninth row. Elements 4411 through 4440 are in the one hundred fiftieth row. Elements 4441 through 4470 are in the one hundred fifty-first row. Elements 4471 through 4500 are in the one hundred fifty-second row. Elements 4501 through 4530 are in the one hundred fifty-third row. Elements 4531 through 4560 are in the one hundred fifty-fourth row. Elements 4561 through 4590 are in the one hundred fifty-fifth row. Elements 4591 through 4620 are in the one hundred fifty-sixth row. Elements 4621 through 4650 are in the one hundred fifty-seventh row. Elements 4651 through 4680 are in the one hundred fifty-eighth row. Elements 4681 through 4710 are in the one hundred fifty-ninth row. Elements 4711 through 4740 are in the one hundred sixtieth row. Elements 4741 through 4770 are in the one hundred sixty-first row. Elements 4771 through 4800 are in the one hundred sixty-second row. Elements 4801 through 4830 are in the one hundred sixty-third row. Elements 4831 through 4860 are in the one hundred sixty-fourth row. Elements 4861 through 4890 are in the one hundred sixty-fifth row. Elements 4891 through 4920 are in the one hundred sixty-sixth row. Elements 4921 through 4950 are in the one hundred sixty-seventh row. Elements 4951 through 4980 are in the one hundred sixty-eighth row. Elements 4981 through 5010 are in the one hundred sixty-ninth row. Elements 5011 through 5040 are in the one hundred seventieth row. Elements 5041 through 5070 are in the one hundred seventy-first row. Elements 5071 through 5100 are in the one hundred seventy-second row. Elements 5101 through 5130 are in the one hundred seventy-third row. Elements 5131 through 5160 are in the one hundred seventy-fourth row. Elements 5161 through 5190 are in the one hundred seventy-fifth row. Elements 5191 through 5220 are in the one hundred seventy-sixth row. Elements 5221 through 5250 are in the one hundred seventy-seventh row. Elements 5251 through 5280 are in the one hundred seventy-eighth row. Elements 5281 through 5310 are in the one hundred seventy-ninth row. Elements 5311 through 5340 are in the one hundred eightieth row. Elements 5341 through 5370 are in the one hundred eighty-first row. Elements 5371 through 5400 are in the one hundred eighty-second row. Elements 5401 through 5430 are in the one hundred eighty-third row. Elements 5431 through 5460 are in the one hundred eighty-fourth row. Elements 5461 through 5490 are in the one hundred eighty-fifth row. Elements 5491 through 5520 are in the one hundred eighty-sixth row. Elements 5521 through 5550 are in the one hundred eighty-seventh row. Elements 5551 through 5580 are in the one hundred eighty-eighth row. Elements 5581 through 5610 are in the one hundred eighty-ninth row. Elements 5611 through 5640 are in the one hundred ninetieth row. Elements 5641 through 5670 are in the one hundred ninety-first row. Elements 5671 through 5700 are in the one hundred ninety-second row. Elements 5701 through 5730 are in the one hundred ninety-third row. Elements 5731 through 5760 are in the one hundred ninety-fourth row. Elements 5761 through 5790 are in the one hundred ninety-fifth row. Elements 5791 through 5820 are in the one hundred ninety-sixth row. Elements 5821 through 5850 are in the one hundred ninety-seventh row. Elements 5851 through 5880 are in the one hundred ninety-eighth row. Elements 5881 through 5910 are in the one

Bonds between atoms, ions, molecules

Bonding in general are attractive and repulsive forces. There is relationship between type of bonding and some physical properties (e.g. melting point). General properties of engineering materials arise from type of bonding and structure.

Primary bonds (strong) :

- metallic bonding,
- ionic bonding,
- covalent bonding.

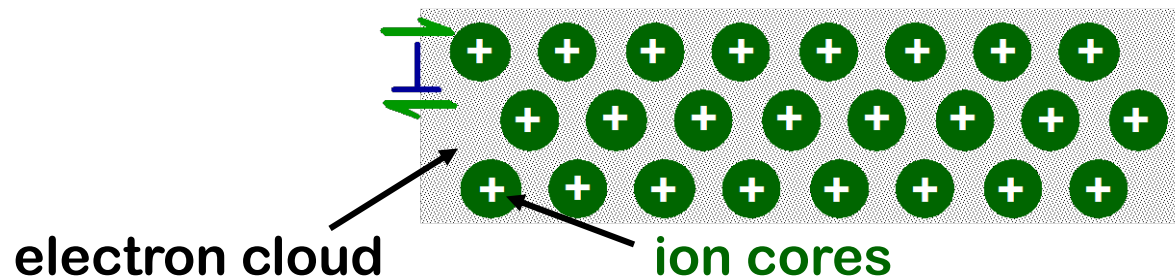
Secondary bonds (weak):

- hydrogen bonding ,
- van der Waals bonding.

Bonds	Bonds energy kJ/mol
Ionic	600-1550
Covalent	500-1250
Metallic	100-850
Van der Waals	< 40

Metallic bonding

- In metallic bonding instead of sharing electrons between two atoms the electrons in the outer shells are shared amongst all the atoms in a lattice with all the atoms positively charge. These atoms are attracted to the negatively charged 'cloud' of electrons.



Metallic bonding

- Outer electrons are delocalized and act as “glue” between positively charged ion cores
- Generally found for elements with one, two or three valence electrons
- Only the outer (valence) electrons participate in the bonding
We distinguish core and valence electrons:
 - core electrons belong to closed shells
 - valence electrons are the outermost electrons, usually in open shells
- Bonding has no directional preference
- Closed-packed atomic configurations are preferred

Physical properties

- Specific Heat – The amount of energy required to raise the temperature of a mass of material 1 degree.
 - Aluminum 902 [J/kg·K]
 - Copper 386
 - Brass 382
 - Steel 473
- Thermal Conductivity is the ease with which heat flows through a material.
 - Copper 400 [W/m·K]
 - Aluminum 236
 - Brass 114
 - Steel 43
 - Stainless steel 14

Physical properties

- Thermal Expansion is the amount a material expands when heated.
 - Aluminum 12.44×10^{-6} [1/K]
 - Copper 9.0×10^{-6}
 - Iron 6.55×10^{-6}
 - Steel 6.55×10^{-6}
 - Titanium 4.0×10^{-6}

Crystalline structure

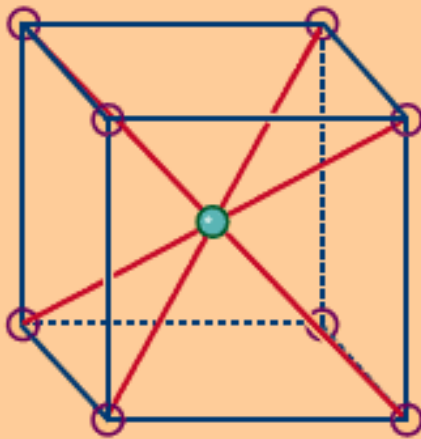
A *unit cell* of the crystal is an imaginary parallel-sided region from which the entire crystal can be built up.

Usually the smallest unit cell which exhibits the greatest symmetry is chosen. If repeated (translated) in 3 dimensions, the entire crystal is formed.

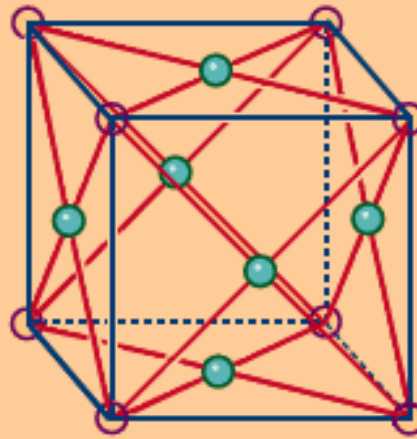
Since metal atoms and ions lack directional bonding, they will often pack with greatest efficiency.

In *close or closest packing*, each metal atom has 12 nearest neighbors. The number of nearest neighbors is called the *coordination number*.

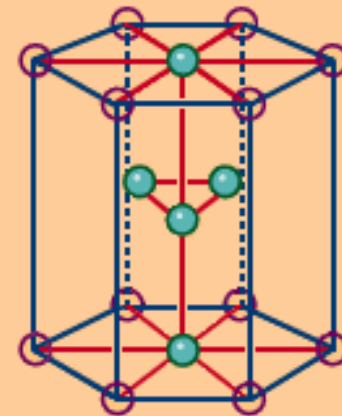
Crystalline structures of metals



bcc



fcc



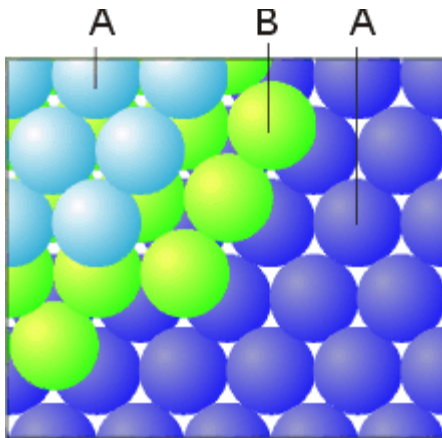
hcp

The body-centred cubic (bcc) structure has atoms in the middle of a cube and at its corners. The face-centred cubic (fcc) structure shows atoms at the centre of each face and at the corners of the cube. In the hexagonal close-packed (hcp) structure, a prism of hexagonal shape is formed when an upper layer of close-packed atoms is superimposed directly over a bottom layer of the same arrangement; an intermediate layer of atoms is pressed closely into the spaces between them.

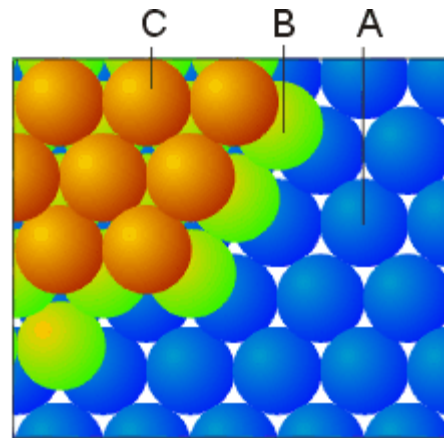
Close packing structures

- Six atoms surround an atom in the same plane, and the central atom is then “capped” by 3 atoms on top, and 3 atoms below it.
- If the bottom “cap” and the top “cap” are directly above each other, in an ABA pattern, the arrangement has a hexagonal unit cell, or is said to be **hexagonal close packed (hcp)**.
- If the bottom and top “caps” are staggered, in an ABC pattern, the unit cell that results is a **face-centered cubic (fcc)**.

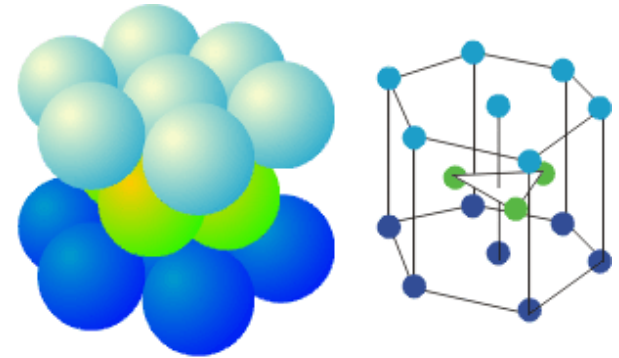
Close packing structures



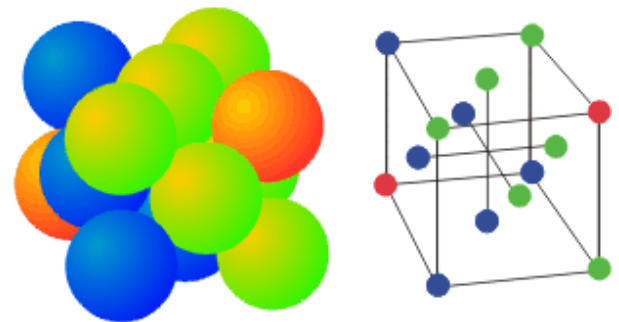
(a)



(b)



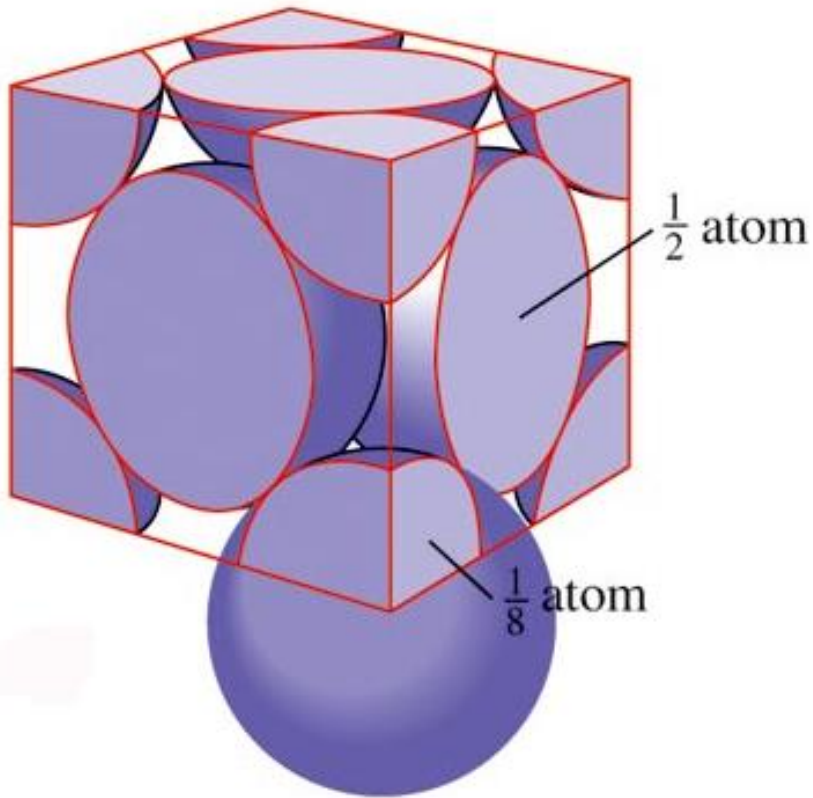
(a)



(b)

Unit cells: a) hexagonal close packed, b) face centered cubic

Crystalline structure

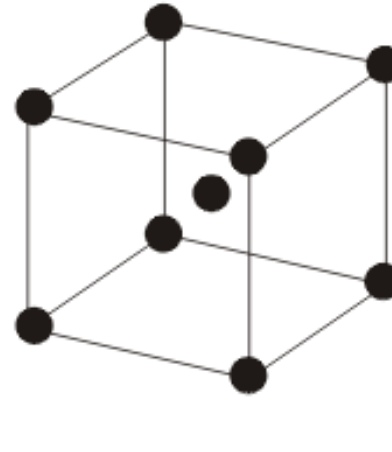
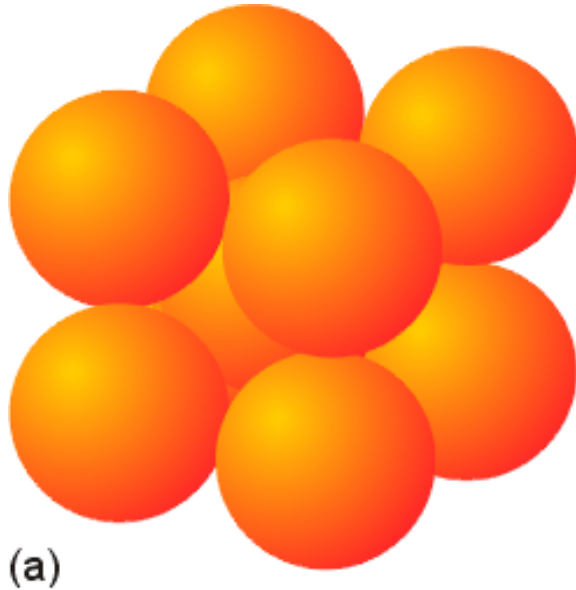


Face-centered cubic (fcc) unit cell contains a total of 4 atoms: 1 from the corners, and 3 from the faces.

Each metal atom has 12 nearest neighbors.

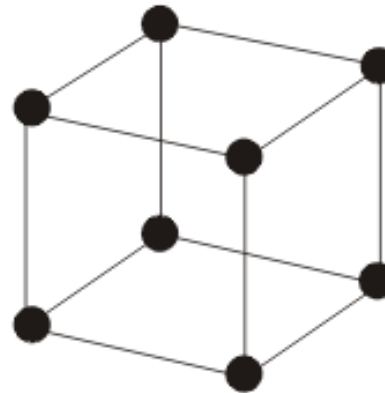
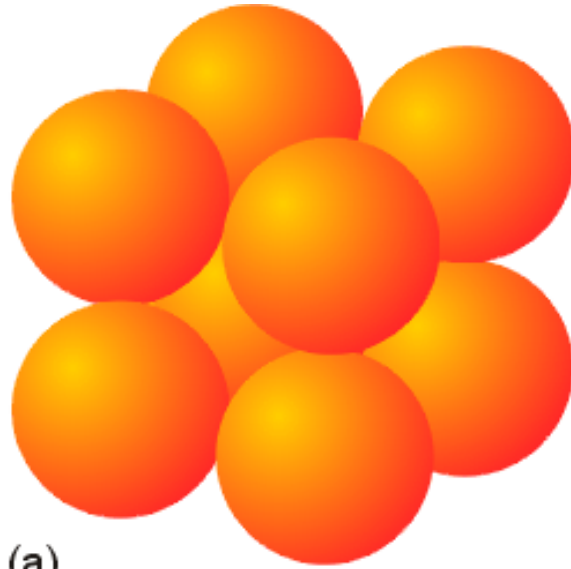
The packing efficiency is 74%.

Crystalline structure



Body-centered cubic (bcc) unit cells have an atom in the center of the cube and one in each corner. The packing efficiency is 68%, and the coordination number = 8.

Crystalline structure



Simple cubic (or primitive cubic) unit cells are relatively rare. The atoms occupy the corners of a cube. The coordination number is 6, and the packing efficiency is only 52.4%.

Allotropy or Polymorphism

Many metals exhibit different crystal structures with changes in pressure and temperature.

Typically, denser forms occur at higher pressures.

Higher temperatures often cause close-packed structures to become body-center cubic structures due to atomic vibrations.

Structure of alloys

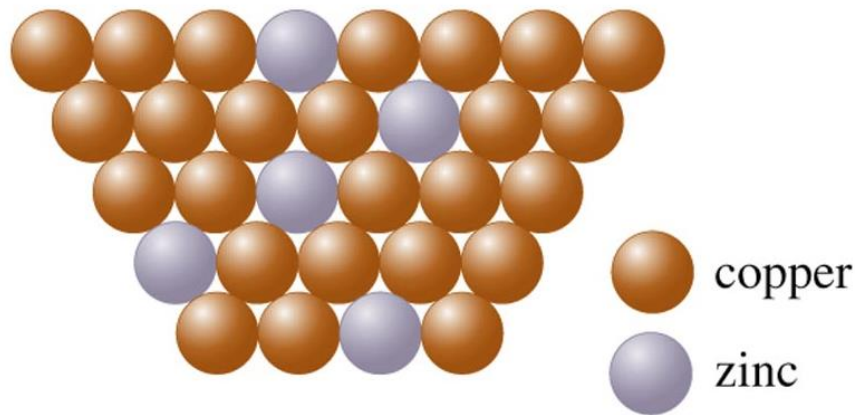
An alloy is a mixture of different metals or metals with nonmetals that often have characteristics superior to any of the apparent elements.

Phases which can form microstructure of alloys:

- Pure metals,
- Solid solutions (substitutional, interstitial),
- Intermediate phases (Intermetallic compounds).

Substitutional solid solution

- Substitutional solid solution has a structure in which sites of the solvent metal are occupied by solute metal atoms. An example is brass (an alloy of zinc and copper).



Brass

Substitutional solid solution

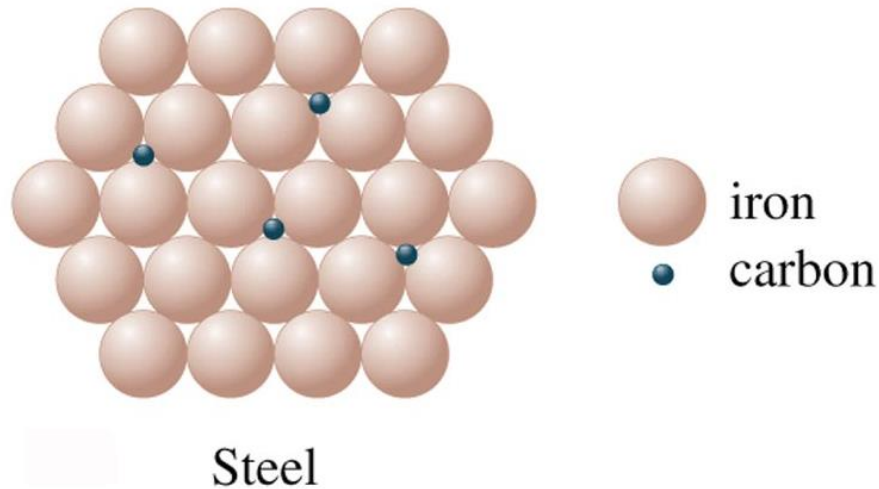
Two elements could have unlimited solubility if:

- the atomic radii of the two metals are within 15% of each other,
- the unit cells of the pure metals are the same,
- the electropositive nature of the metals is similar (to prevent a redox reaction),

Interstitial solid solution

Interstitial alloys are solid solutions in which the solute atoms occupy holes (interstices) within the solvent metal structure. An example is steel (an alloy of iron and carbon).

These solutions often have a non-metallic solute that will fit in the small holes of the metal lattice.



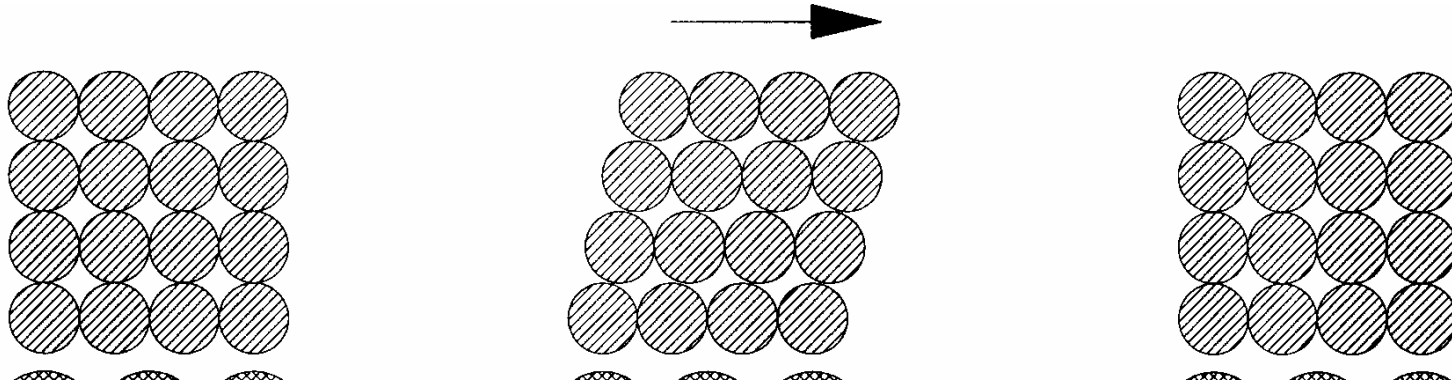
Intermediate phases

- There are usually limits to the solubility of one element in another
- When the amount of the dissolving element in the alloy exceeds the solid solubility limit of the base metal, a second phase forms in the alloy
- The term *intermediate phase* is used to describe it because its chemical composition is intermediate between the two pure elements
- Its crystalline structure is also different from those of the pure metals

Types of intermediate phases

1. *Metallic compounds* – consist of a metal and nonmetal, such as Fe_3C
2. *Intermetallic compounds* - two metals that form a compound, such as Mg_2Pb
 - In some alloy compositions, the intermediate phase is mixed with the primary solid solution to form a two-phase structure
 - Some two-phase alloys are important because they can be heat treated for much higher strength than solid solutions

Elastic deformation



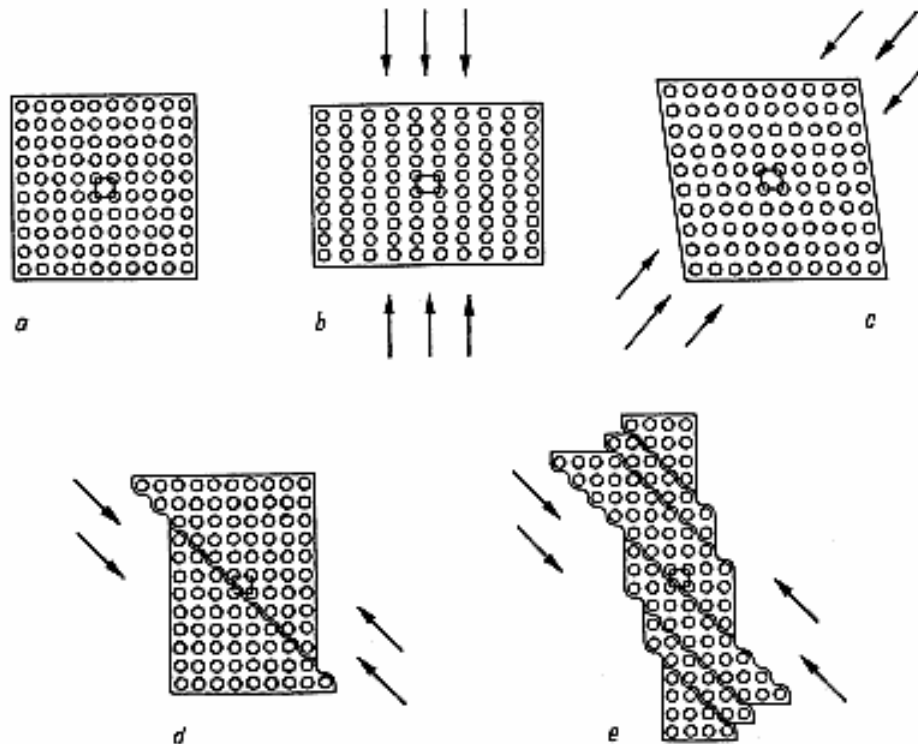
- When a sufficient load is applied to a metal or other structural material, it will cause the material to change shape. This change in shape is called **deformation**.
- Elastic deformation describes the effect of a metal returning to its original shape when a load (or force) is removed.

Plastic deformation

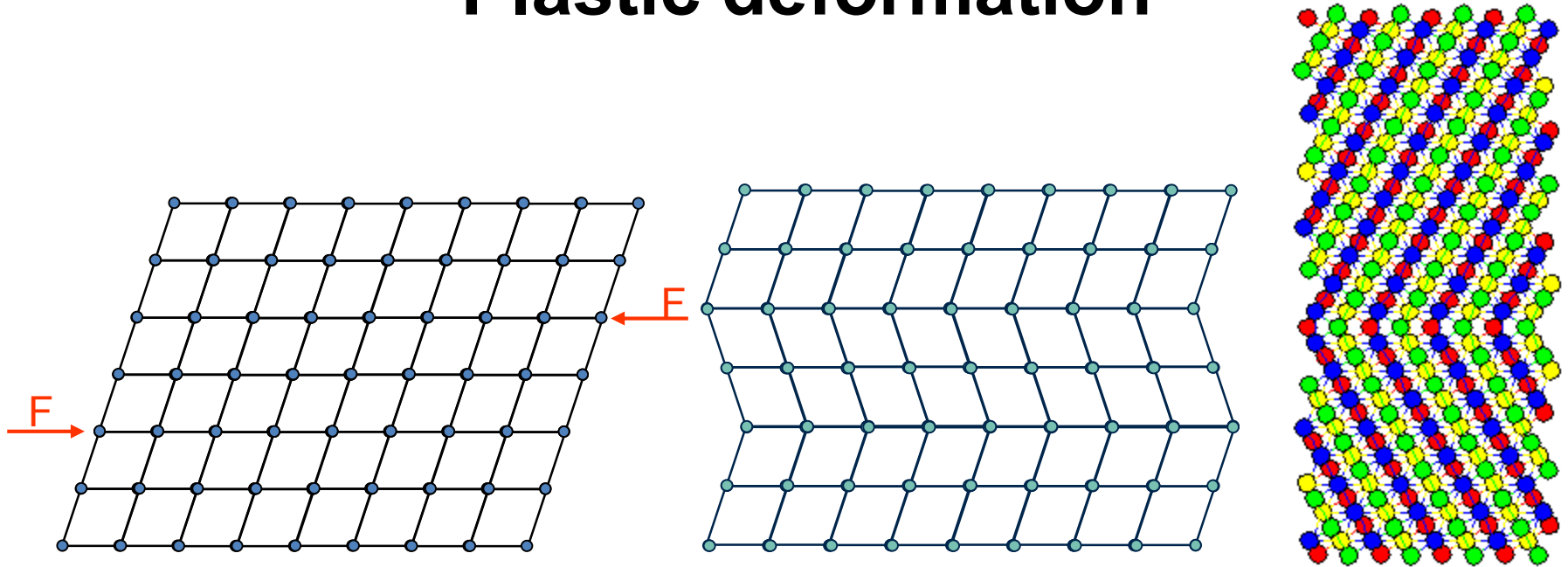
Plastic deformation results when stress is so great the form is permanently altered. Plastic deformation is always preceded by elastic deformation.

Two mechanisms of plastic deformation:

- slip,
- twinning.

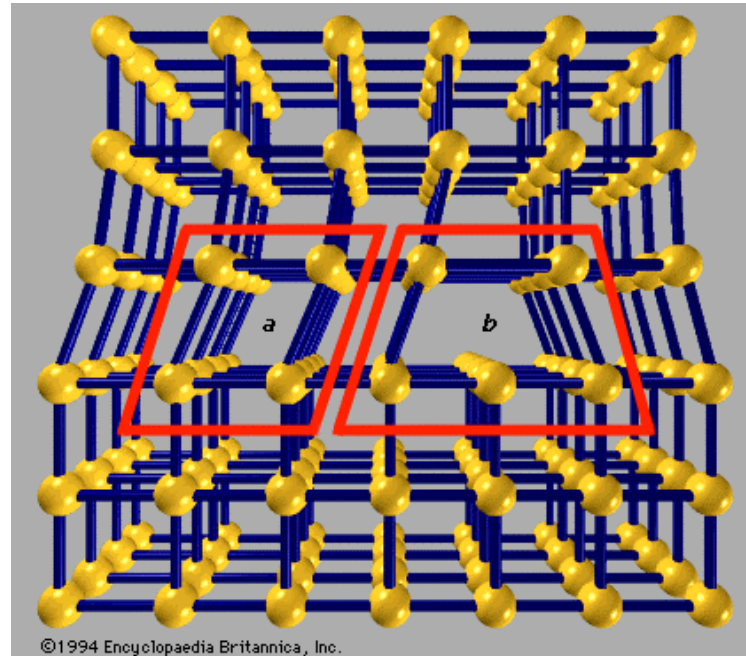


Plastic deformation



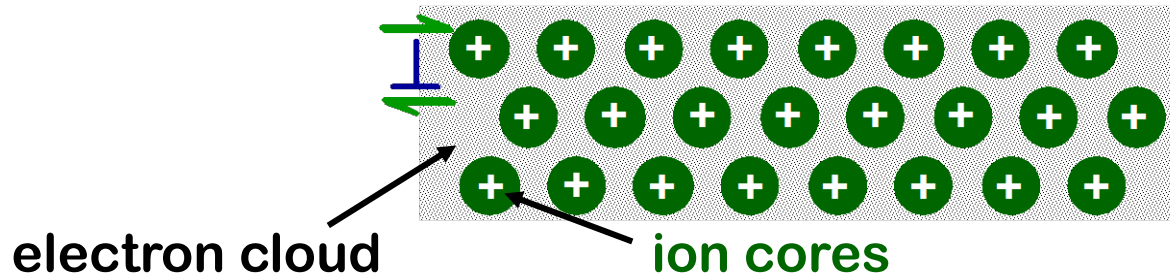
Twinning - a shear force can produce atomic displacements so that on one side of the plane (the twin boundary), atoms are located in mirror image positions to atoms on the other side.

Plastic deformation



In edge dislocations, distortion exists along an extra half-plane of atoms. These atoms also define the dislocation line. Motion of many of these dislocations will result in plastic deformation. Edge dislocations move in response to shear stress applied perpendicular to the dislocation line.

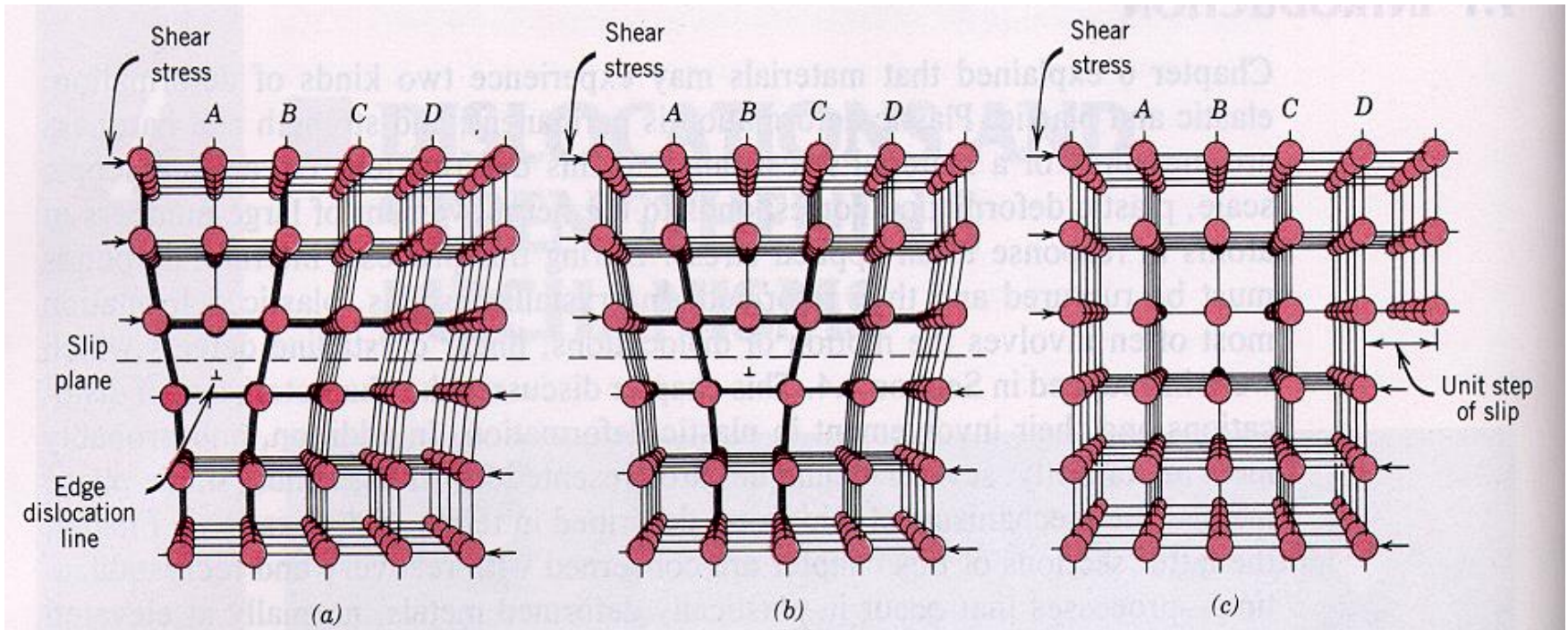
Plastic deformation



Dislocations motion in metals is easier than in other materials (ceramics) due to:

- non directional bonding,
- close packed structure and directions facilitating slip.

Plastic deformation

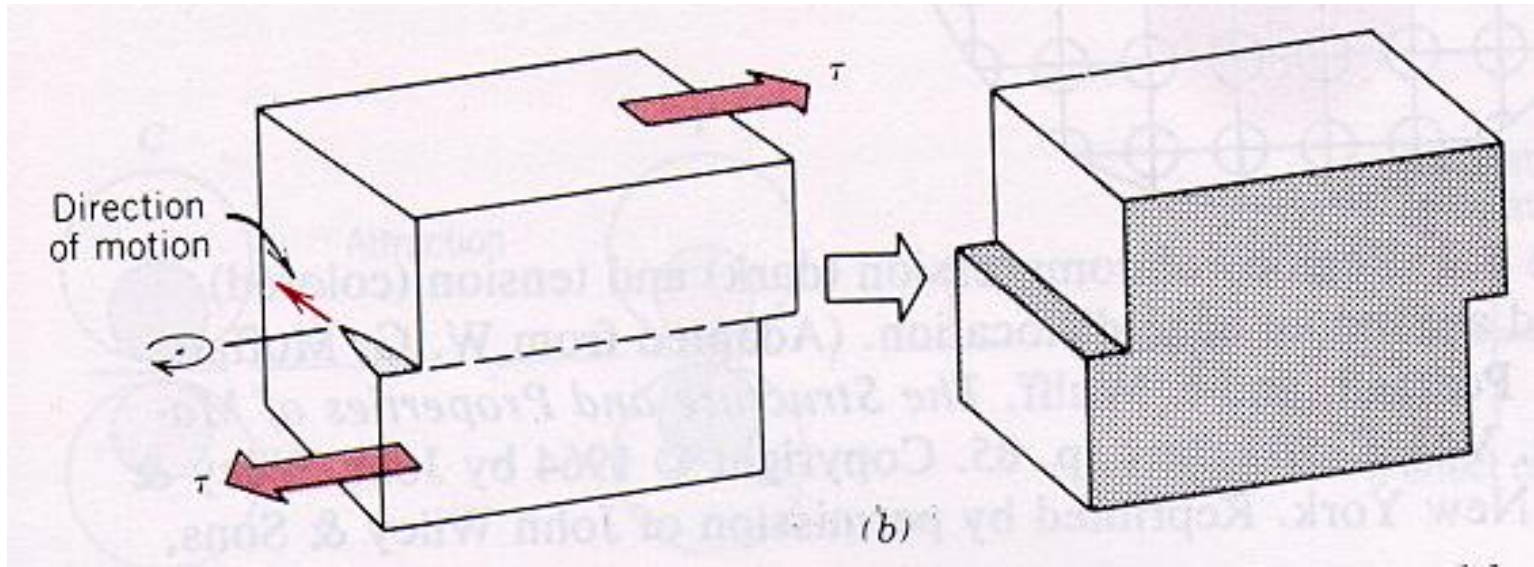


As the dislocation moves, the extra half plane will break its existing bonds and form new bonds with its neighbor opposite of the dislocation motion.

This step is repeated in many discrete steps until the dislocation has moved entirely through the lattice.

After all deformation, the extra half plane forms an edge that is one unit step wide

Plastic deformation

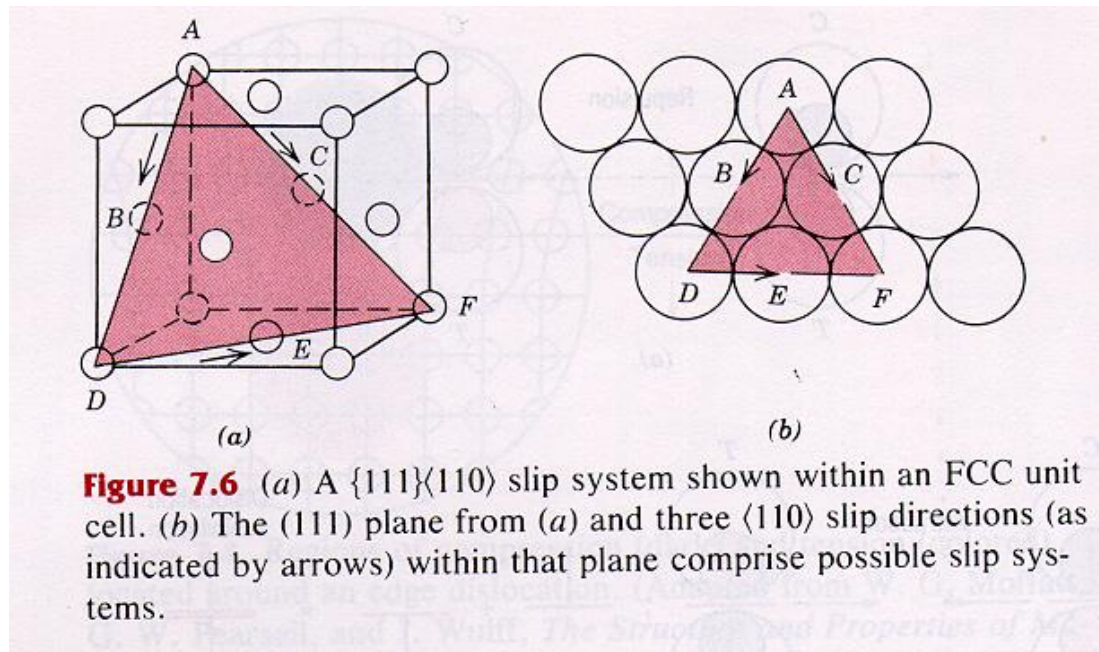


The motion of a screw dislocation is also a result of shear stress. Motion is perpendicular to direction of stress, rather than parallel (edge). However, the net plastic deformation of both edge and screw dislocations is the same. Most dislocations can exhibit both edge and screw characteristics. These are called mixed dislocations.

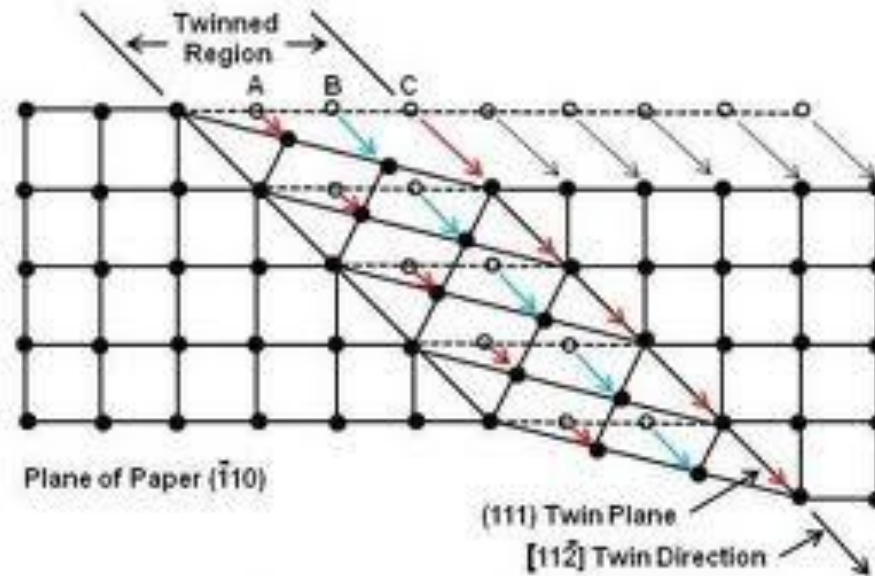
Plastic deformation

Usually there are preferred slip planes and directions in certain crystal systems. The combination of both the slip plane and direction form the **slip system**.

Slip plane is generally taken as the closest packed plane in the system
Slip direction is taken as the direction on the slip plane with the highest linear density.



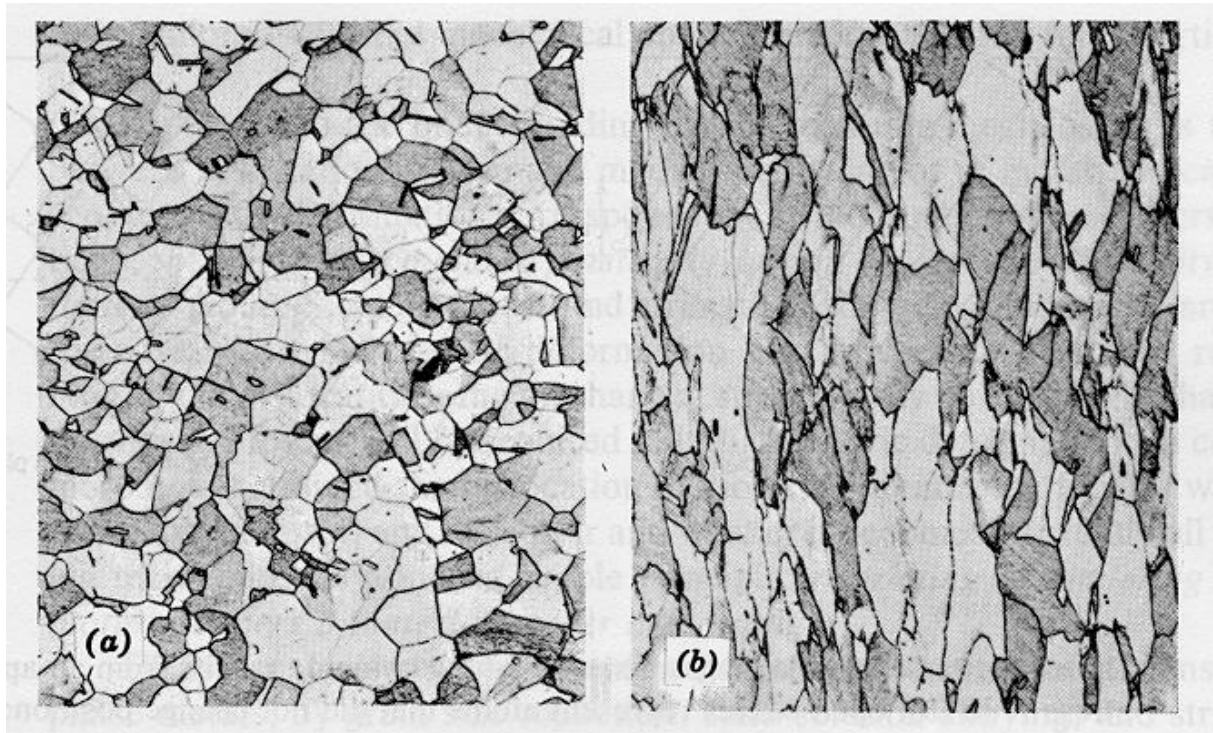
Plastic deformation



Schematic Diagram of Twinning in an f.c.c. Lattice

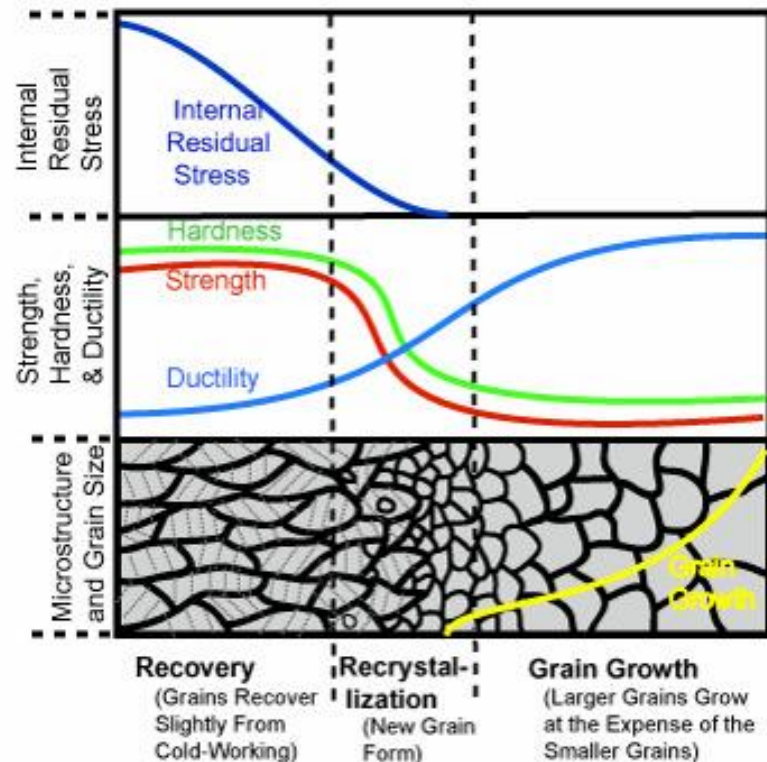
Plastic deformation

Work hardening is when a metal is strained beyond the yield point. An increasing stress is required to produce additional plastic deformation and the metal apparently becomes stronger and more difficult to deform.



Recrystallization

- At a higher temperature, new, strain-free grains nucleate and grow inside the old distorted grains and at the grain boundaries.
- These new grains grow to replace the deformed grains produced by the strain hardening.
- With **recrystallization**, the mechanical properties return to their original weaker and more ductile states.

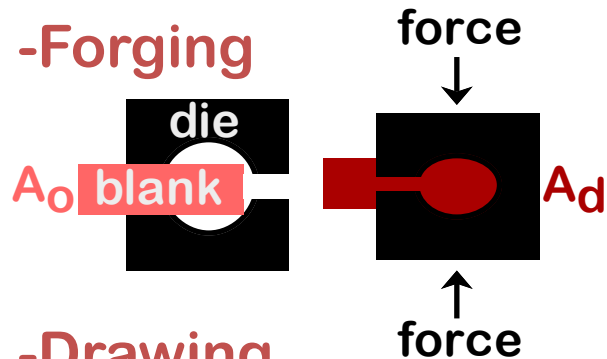


Cold work

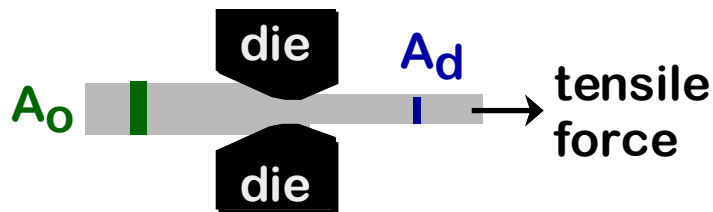
$$\%CW = \frac{A_o - A_d}{A_o} \times 100$$

- Room temperature deformation.
- Common forming operations change the cross sectional area:

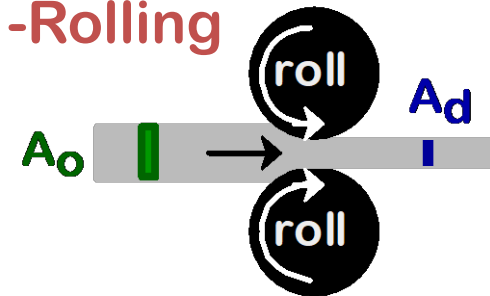
-Forging



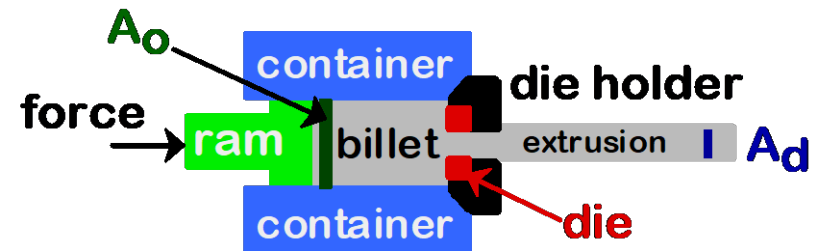
-Drawing



-Rolling



-Extrusion



Classification of metals and alloys

Metals are available in **pure** or **alloy** form.

Pure metals such as pure aluminium or pure copper, contain only one type of metal. They are not mixed with any other metal.

Pure metals are very soft and therefore rarely used in engineering applications.

Alloys: metal mixed with one or more other elements (metallic or not)

Alloys tend to have better strength properties than pure metals.

Alloys and pure metals often have special physical properties.

Classification of metals and alloys

Ferrous materials - iron as the base metal – steels and cast irons. Steels range from plain carbon (>98% Fe) to high alloy steel (<50% alloying elements).

Nonferrous materials consist of the rest of the metals and alloys: aluminum, magnesium, copper, nickel, titanium, zinc, lead, tin, molybdenum, tungsten, gold, silver, platinum, and others.

Steel - an iron-carbon alloy containing from 0.02% to 2.1% carbon

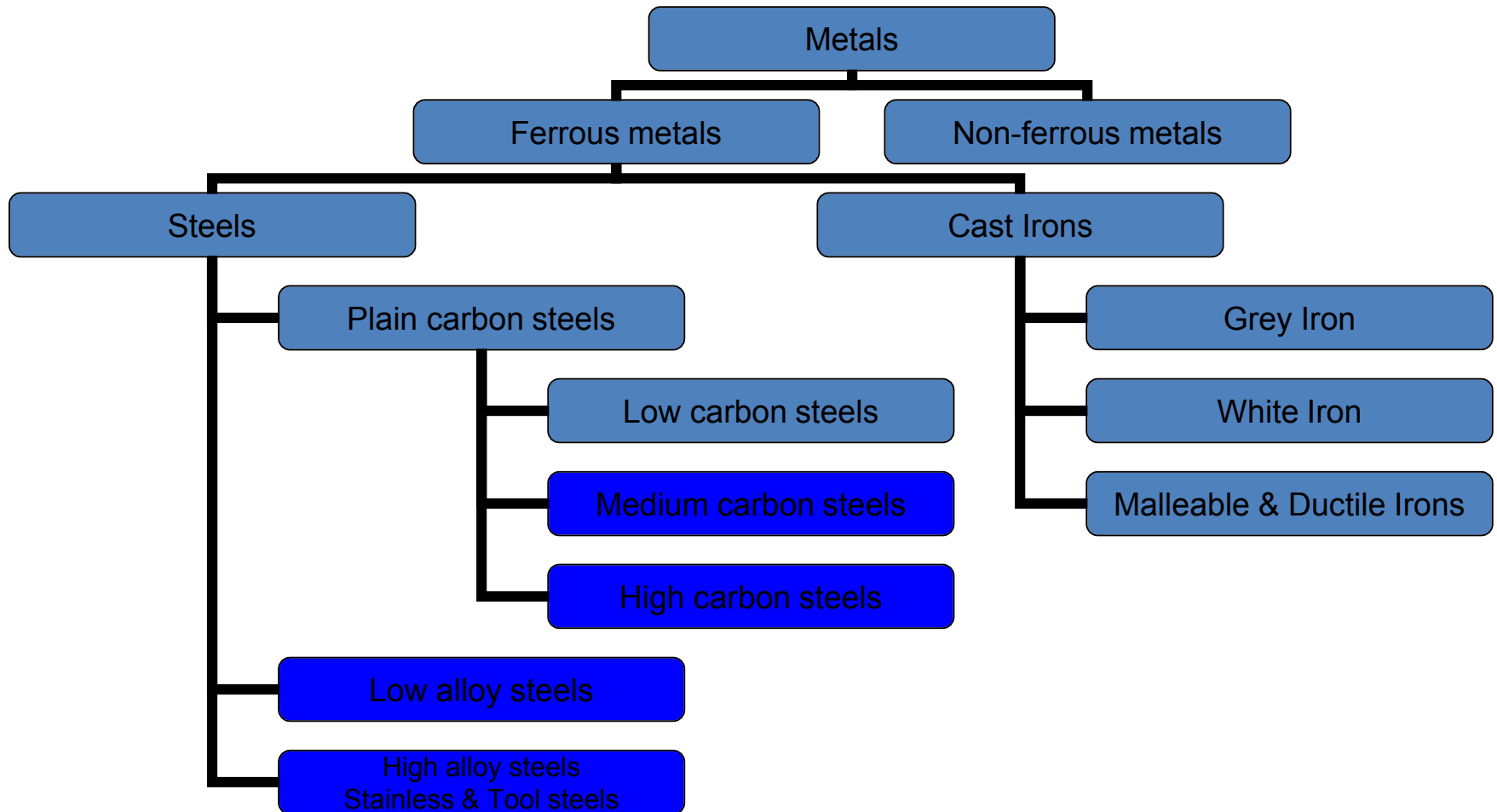
Cast iron - an iron-carbon alloy containing from 2.1% to about 4% or 5% carbon

Steels and cast irons can also contain other alloying elements besides carbon

Within each group of alloy, classification can be made according to:

- (a) chemical composition, e.g. carbon content or alloys content in steels,
- (b) finished method, e.g. hot rolled or cold rolled,
- (c) product form, e.g. bar, plate, sheet, tubing, structural shape,
- (d) method of production, e.g. cast, wrought alloys.

Classification of metals and alloys



Classification of steels

- AISI (American iron and Steel Insitute) numbering system is used in classifying steels.
- SAE (Society of Automotive Engineers).
- 4 digit number: the first two numbers are the type of steel and the last two are the percent of carbon.

Classification of steels

First two numbers

Type of Steel

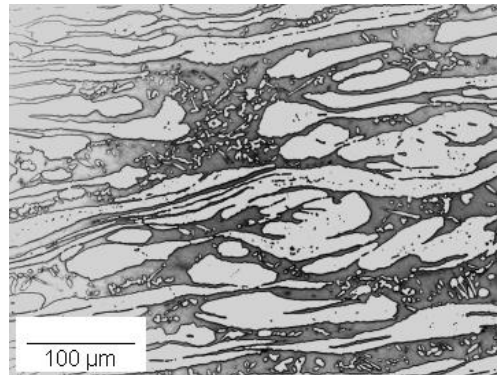
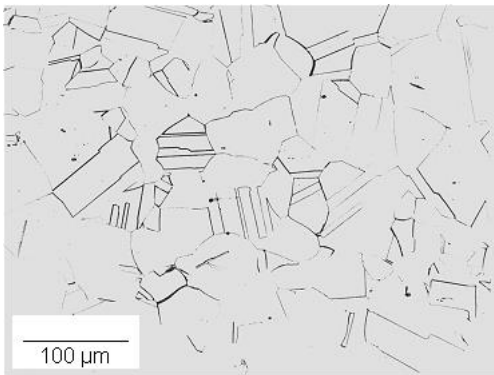
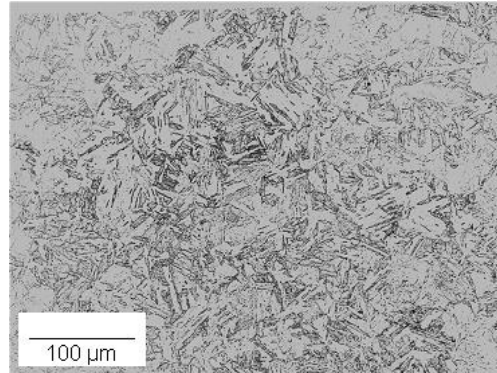
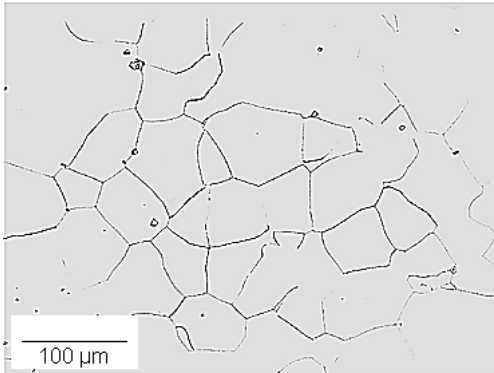
- | | |
|--|---------------------------------|
| – 10 | Carbon Mn to 1% Max |
| – 11 | Resulfurized |
| – 12 | Resulfurized & Rephosphorized |
| – 15 | Plain carbon Mn range 1 - 1.65% |
| – 13 | Manganese Steel Mn 1.75% |
| – 23 | Nickel Steel up to 3.5% |
| – 25 | Nickel Steel up to 5.00% |
| – 31 | Ni 1.25% Cr 0.65-0.80% |
| – 32 | Ni 1.75% Cr 1.07% |
| – 33 | Ni 3.50% Cr 1.50-1.57% |
| – 34 | Ni 3.00% Cr 0.77% |
| – 40 | Molybdenum Steel Mo 0.2-0.25% |
| – 44 | Molybdenum Steel Mo 0.4-0.52% |
| – 41 | Chromium-Molybdenum |
| – 43, 47, 81, 86, 87, 88, 93, 94, 97, 98 | Nickel Chromium-Molybdenum |
| – 50, 51, 50xxx, 51xxx, 52xxx | Chromium Steels |

Stainless steel

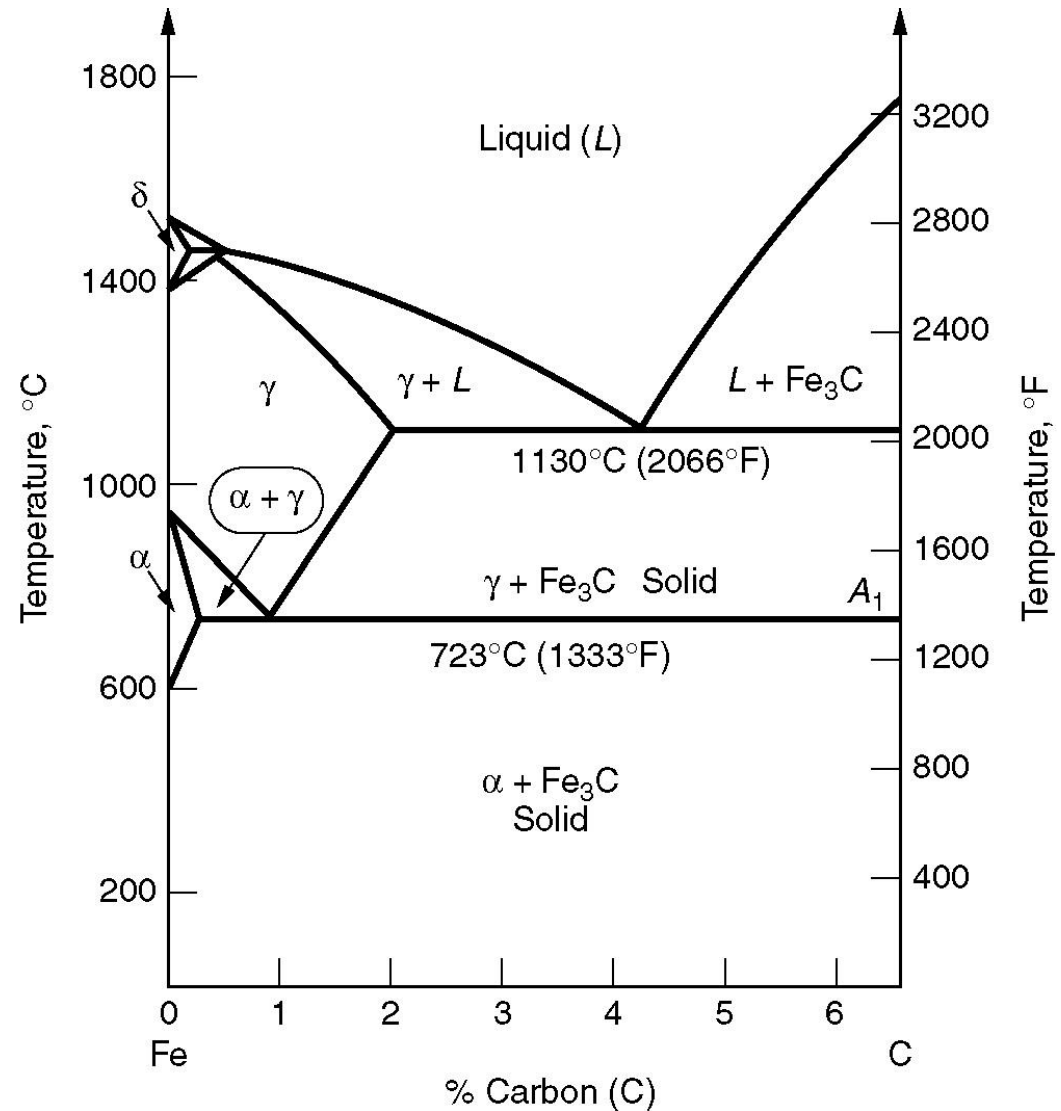
Plain carbon steels rust, and oxidise if heated in air

Adding Cr $\rightarrow$ Cr_2O_3 formation, an oxide which protects the underlying alloy

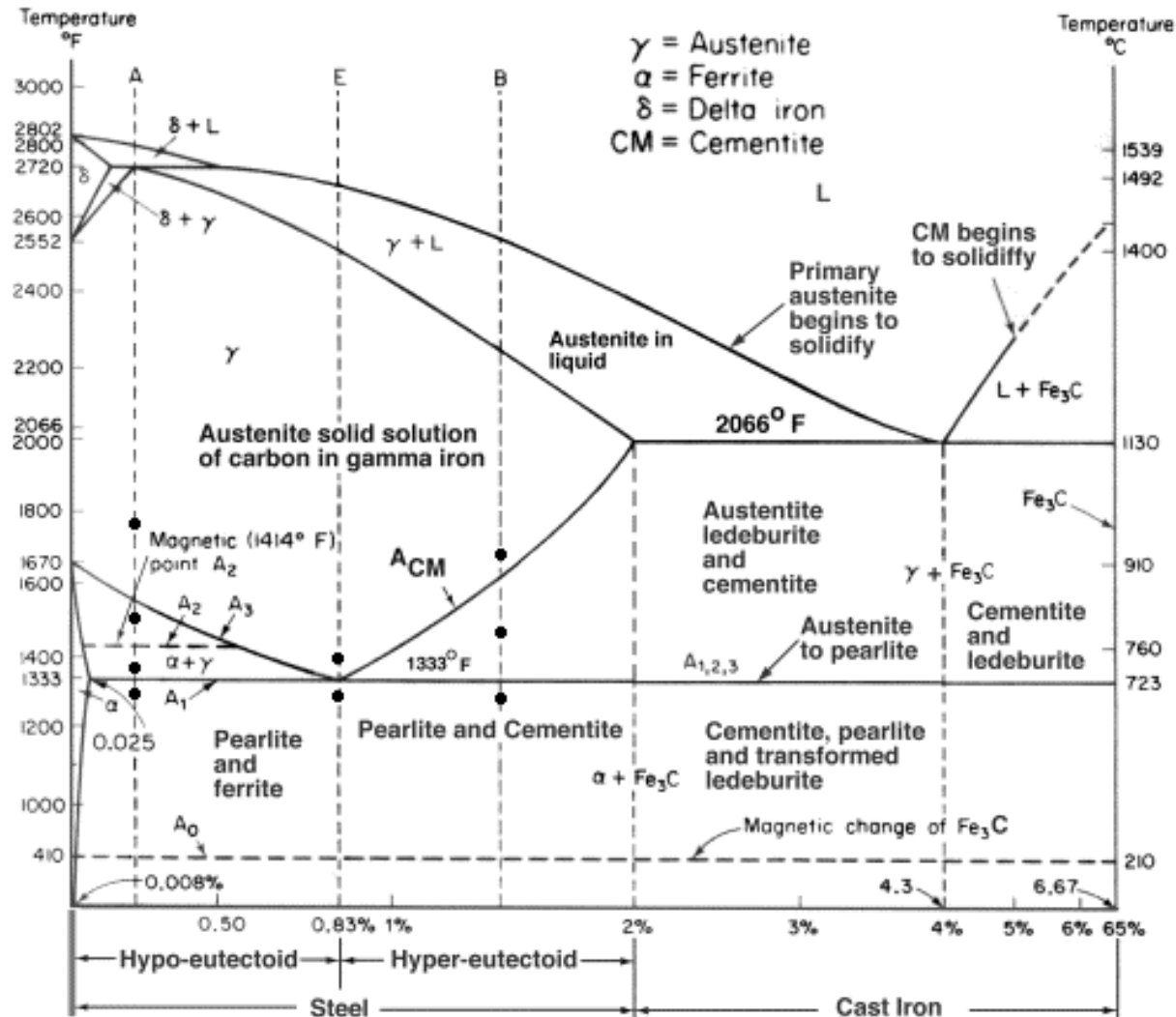
Definition: Fe with >11 wt% Cr = stainless steel



Iron-carbon phase diagram



Iron-carbon phase diagram



Cast irons

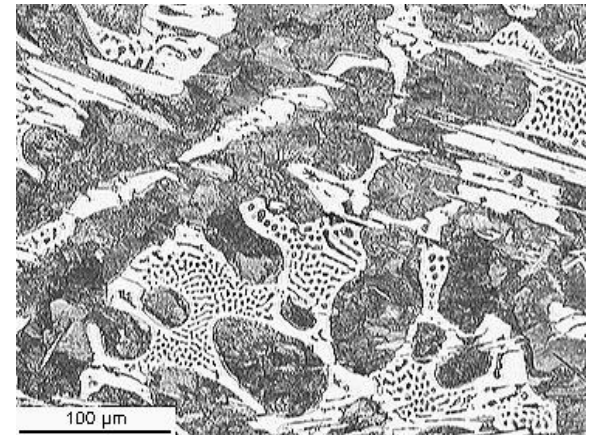
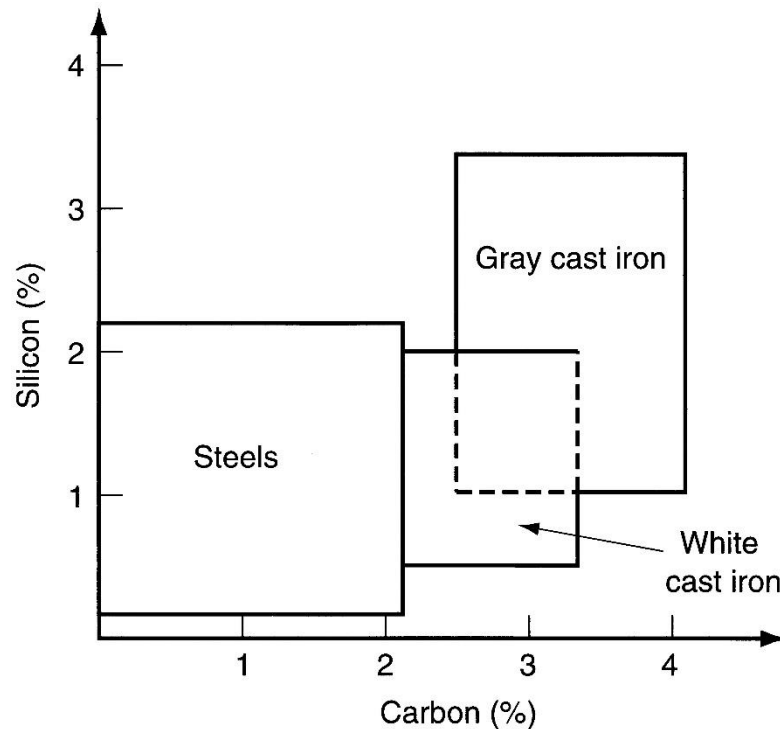
Two types:

White cast iron: $\text{Fe} + \text{Fe}_3\text{C}$

Grey cast iron: $\text{Fe} + \text{C}$ (graphite)

- *Grey iron - formation of graphite rather than cementite promoted through high C and Si content, slow solidification rate.*
- *Ductile (nodular) and malleable irons possess chemistries similar to the gray and white cast irons, respectively, but result from special processing treatments.*
 - Ductile (nodular) iron is formed by special melting and pouring treatment (formed by adding Ce or Mg) of gray cast iron,
 - Malleable iron is formed by heat treatment of white cast iron.

Cast irons



White cast irons

- microstructure based on Fe + Fe₃C

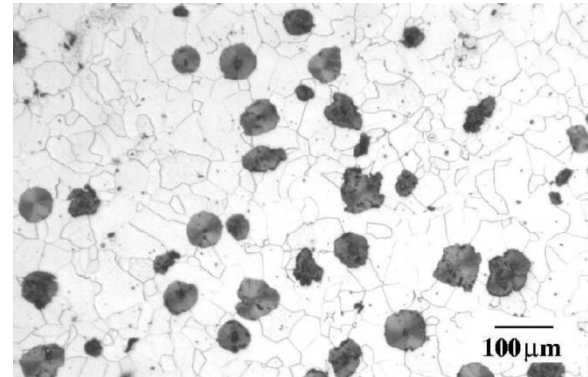
Grey cast irons

- silicon (Si) added 1-3,5%
- Si stabilises the Fe-C system, rather than Fe-Fe₃C, so microstructure is based on Fe + C (free graphite)

Cast irons

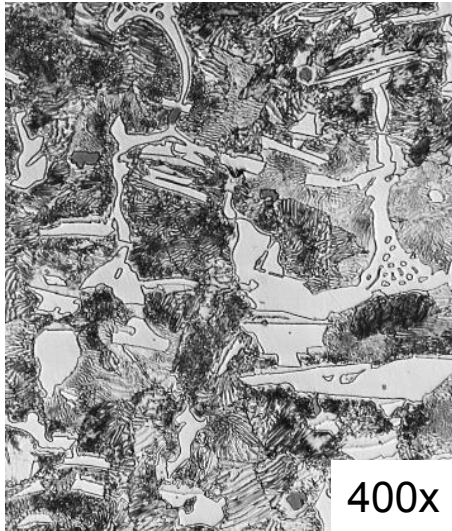
The properties of grey cast iron are strongly dependant on the shape of the graphite.

- flakes
- spheroids - relatively high toughness and ductility, higher strength



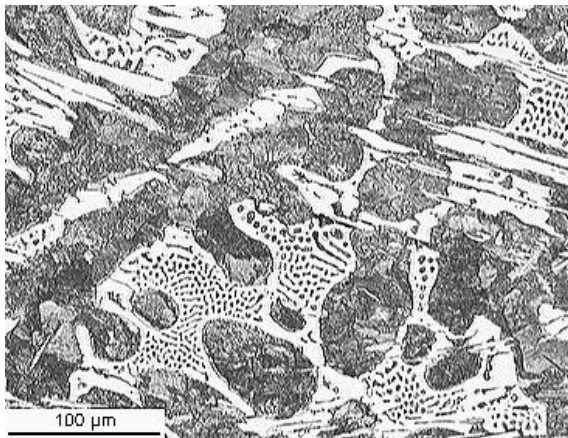
Cast irons

White cast iron



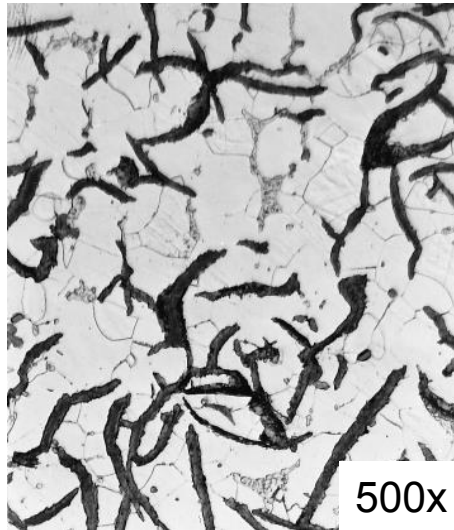
Combination of cementite and pearlite makes white cast iron extremely hard & brittle

Applications limit to e.g. as rollers in rolling mills, and intermediary material for malleable cast iron



Cast irons

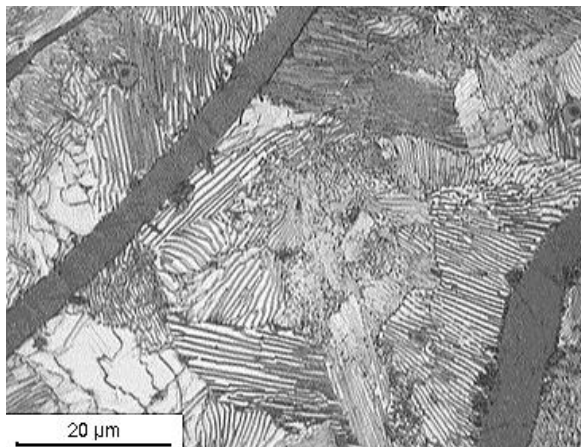
Grey cast iron



Tips of graphite flakes act as points of stress concentration, thus makes grey cast irons weak and brittle in tension

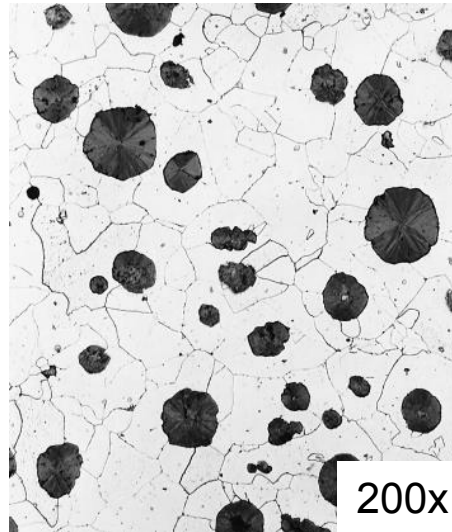
But used extensively due to:

- effective in damping
- high resistance to wear
- easy casting
- low cost



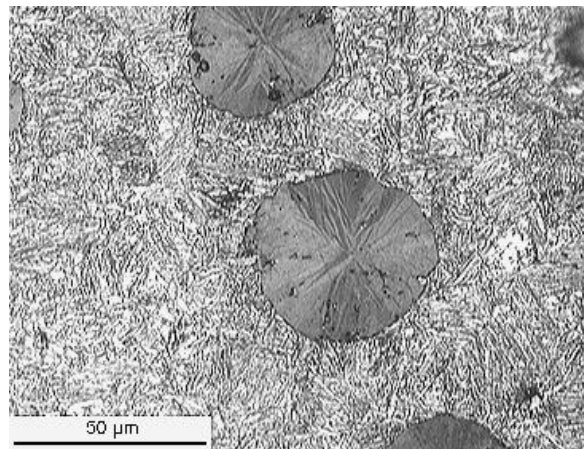
Cast irons

Ductile (nodular) cast iron



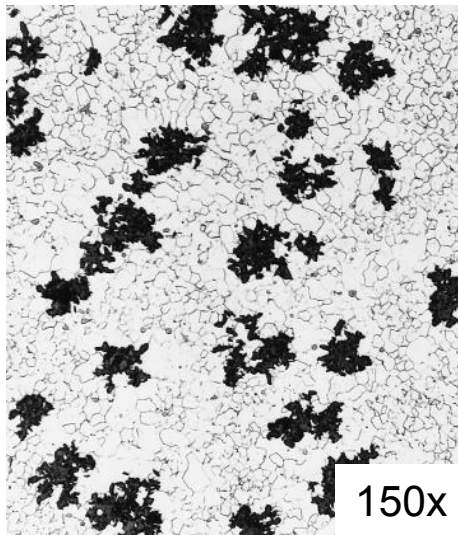
Graphite nodules makes nodular cast iron strong and ductile, approaching carbon steels

Applications include valves, pump bodies, crank shafts and gears.



Cast irons

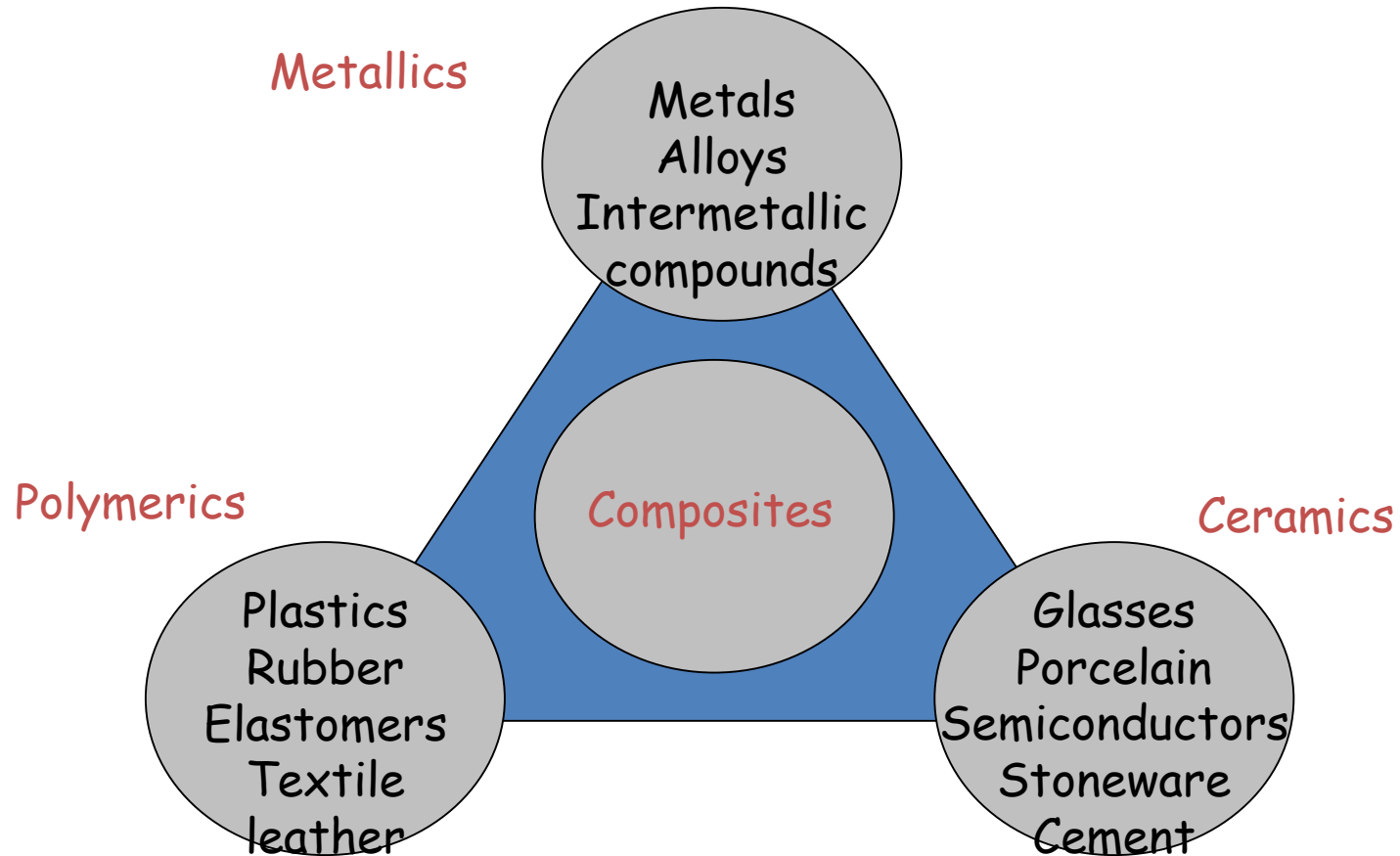
Malleable cast iron



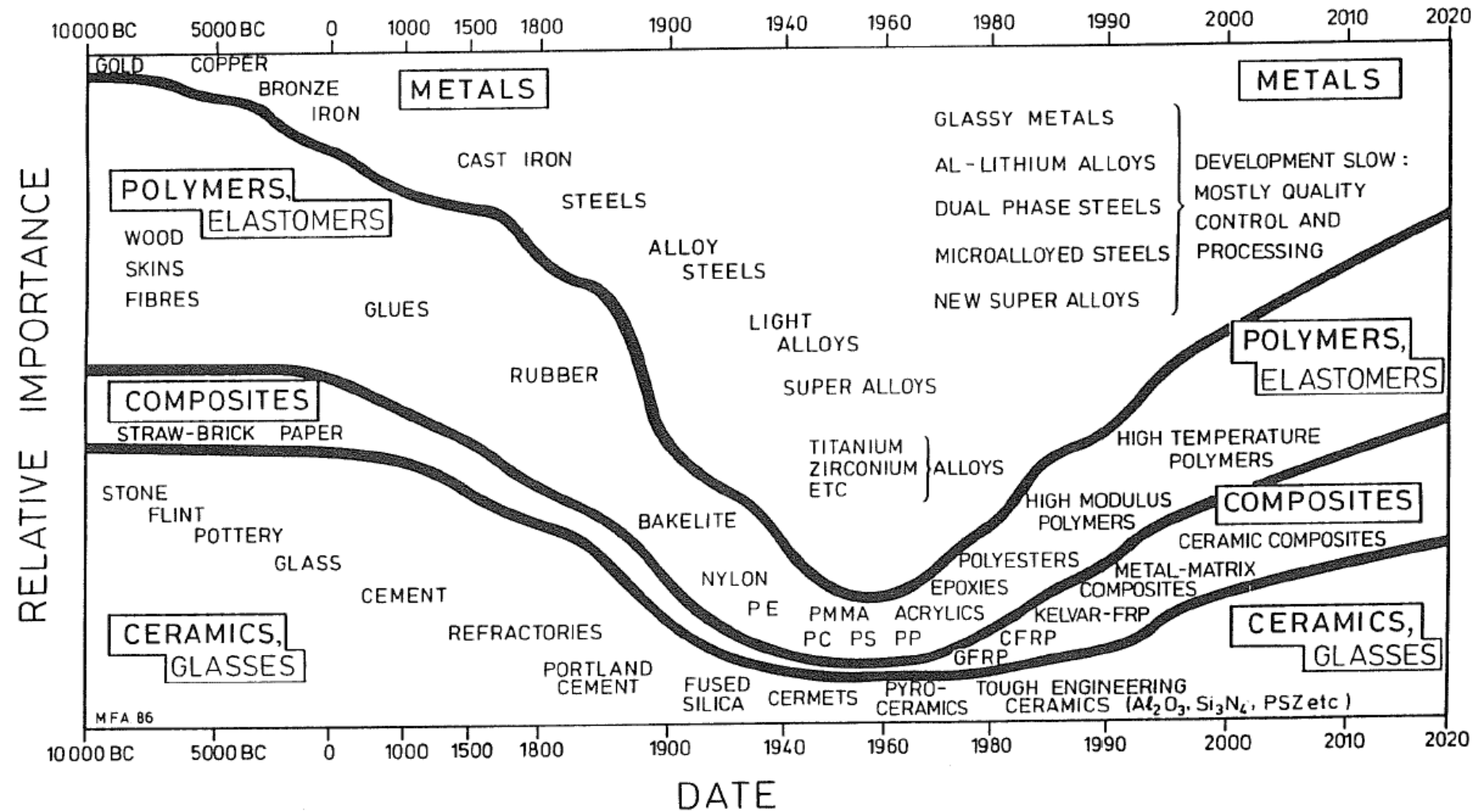
The formation of graphite rosettes gives malleable cast iron relatively high strength with ductility

Applications include transmission gears, flanges pipe fitting for heavy duty services.

Ceramics



Ceramics



Ceramics and glasses

Ceramics

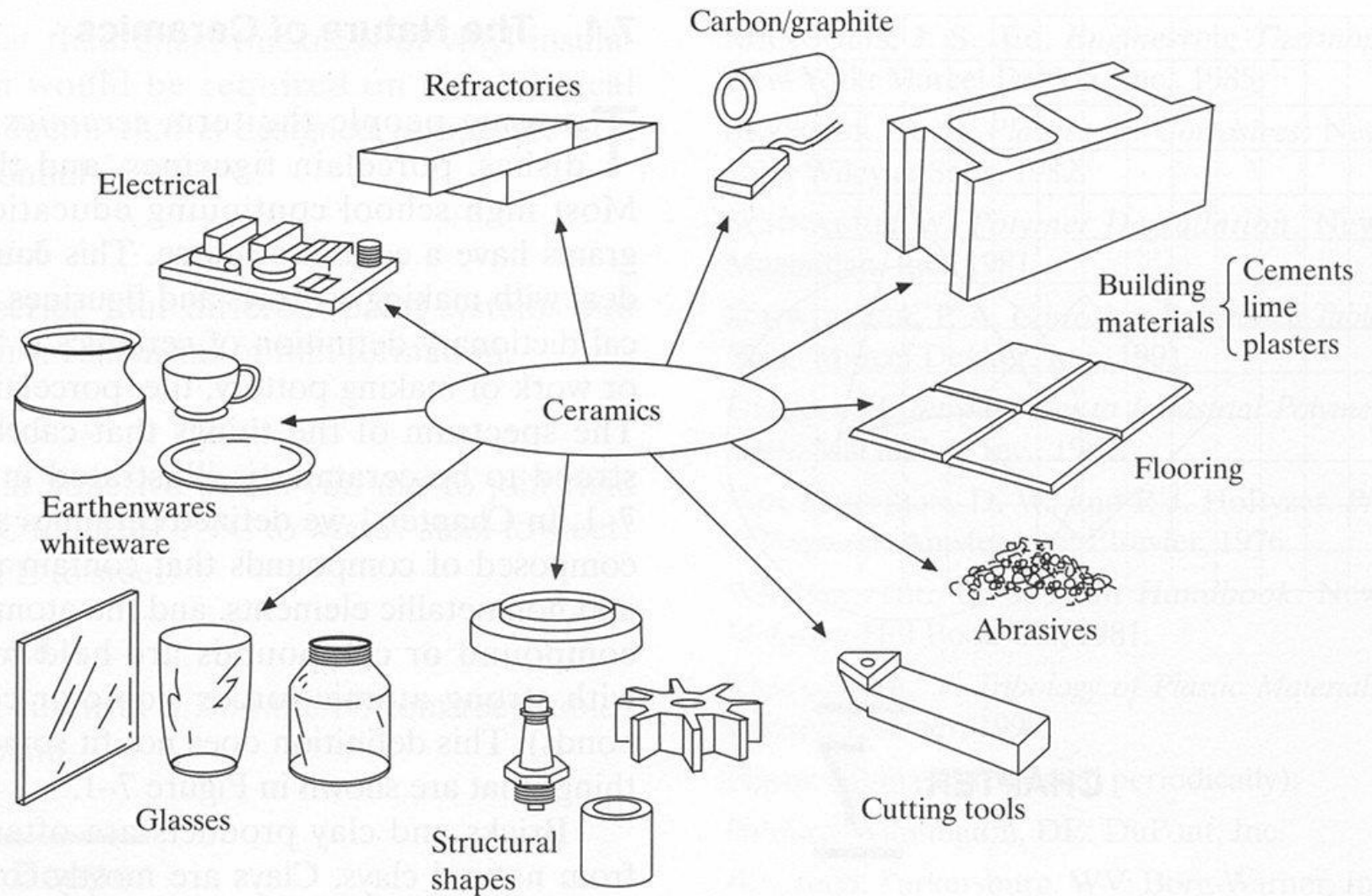
- Inorganic, nonmetallic crystalline materials

Glasses

Inorganic, nonmetallic, material without crystalline structure

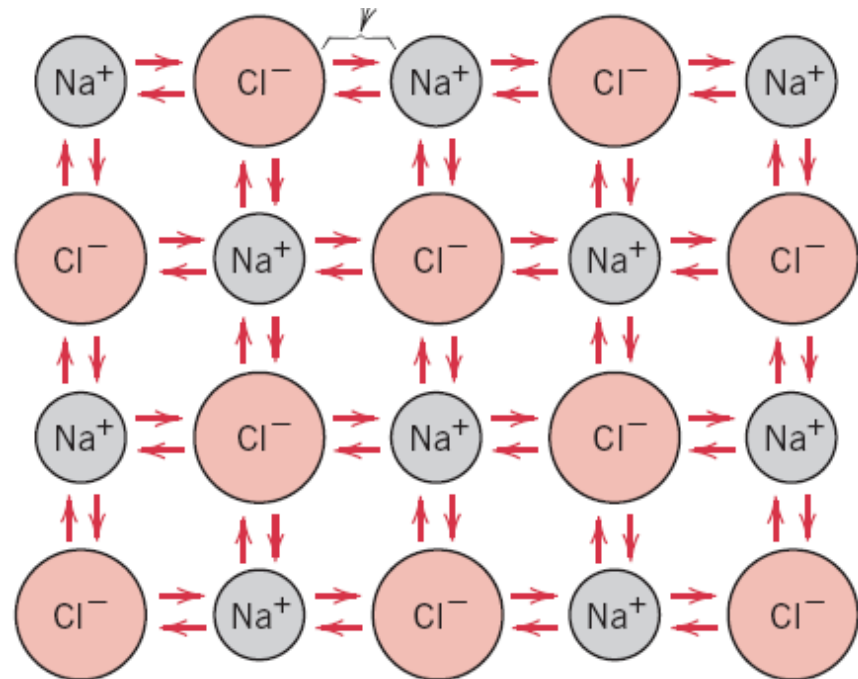
- Traditionally low tech uses (bottles, windows)
- Emerging high tech as semiconductor (silica) and fiber optics

Ceramics



Ceramics

Ceramic materials are compounds of metallic and nonmetallic elements with ionic or covalent bonds, mostly **oxides, nitrides, and carbides**. Ceramic structures do not tolerate plastic deformation mechanisms that occur in metals. They are generally hard and brittle. The ionic bonds are strong, making them high temperature resistant and corrosion resistant. There are no free electrons in the structure, thus they are electrically insulative.



Ceramics

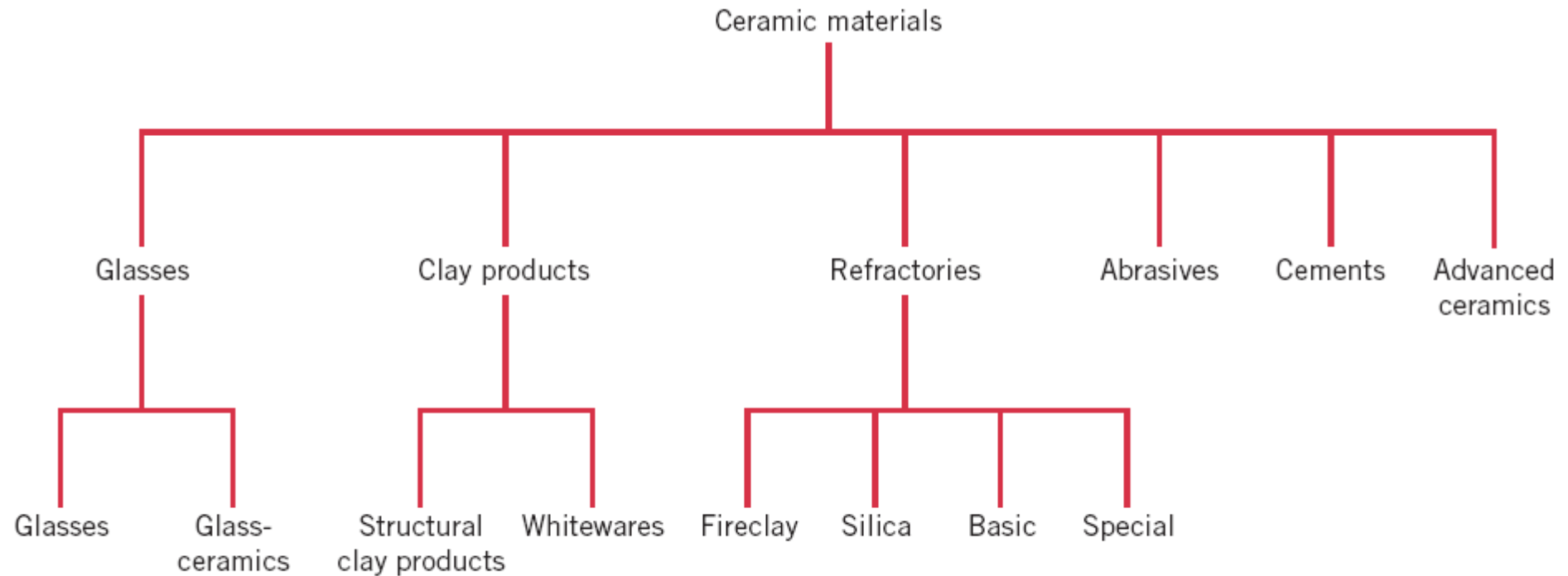
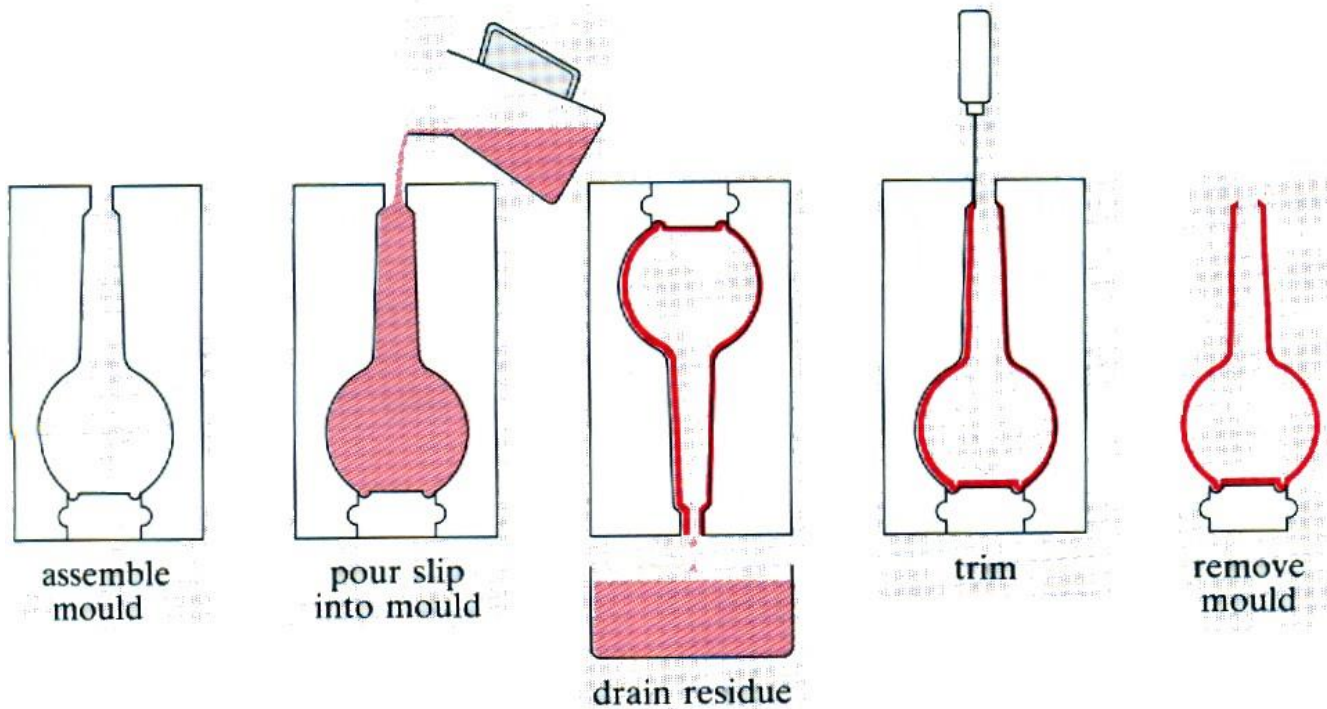


FIGURE 14.1 Classification of ceramic materials on the basis of application.

- Natural ceramics: rock, bricks and tiles, pottery, porcelain, glasses, clayware, lime, plasters
- Industrial ceramics: cement, sewerage pipes, abrasives, refractories
- Advanced ceramics: ceramic components, cutting tools

Ceramics

Ceramics are generally made by taking mixtures of clay, earthen elements, powders, and water and shaping them into desired forms. Once the ceramic has been shaped, it is fired in a high temperature oven known as a kiln. Often, ceramics are covered in decorative, waterproof, paint-like substances known as glazes.



Ceramics

Typical Processing Techniques for Ceramics:

Low temperature setting (chemical reaction) – concrete

Melting and moulding – glasses

Paste (slip) moulding and firing – bricks, tiles, pottery, earthenware, toolware, parts.....

Sintering

Hot isostatic pressing (HIP)

Vitrification (mixing with glass and firing)

Most ceramics are hard and brittle; thus parts cannot be easily machined from standard shapes. Forming ceramics into special shapes by moulding usually involves fabrication of tooling for that particular part

Ceramics

Common Engineering Ceramics

Class	Examples	Typical Uses
Oxides	Alumina (Al_2O_3) Zirconia (ZrO_2) Titania (TiO_2) Silica (SiO_2) Magnesia (MgO) Chromia (Cr_2O_3) Kaolinite ($\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$) Partially stabilised zirconia ($\text{ZrO}_2 - \text{MgO}, \text{CaO}$)	insulator, abrasives, structural parts, refractories Thermal insulation, diamond substitute, jewelry Pigments Abrasive, glass Wear parts, refractories Wear coatings Clay products (bricks, tiles, porcelain.....) Engine parts, structural parts
Carbides	Silicon carbide (SiC) Titanium carbide (TiC) Boron carbide (B_4C) Chromium carbide (Cr_3C_2) Tungsten carbide (WC) Vanadium carbide (VC) Tantalum carbide (TaC)	Abrasives Wear-resistant applications Abrasives Wear-resistant coatings Cutting tools Wear-resistant applications Wear-resistant applications
Nitrides	Silicon nitride (Si_3N_4) Boron nitride (BN)	Wear parts, structural parts, cutting tools, refractories, extrusion dies, pump parts, engine parts, bearings Insulators

Ceramics

Alumina (Al_2O_3):

very good resistance to corrosion and oxidation,
excellent abrasive, also a popular material for electrical insulators

Silicon carbide (SiC):

very good resistance to thermal shock
excellent abrasive

Silicon nitride (Si_3N_4):

very good strength on compression
very good resistance to thermal shock

Zirconia (ZrO_2):

relatively tough, replace steel parts in engines, thermal insulation

Sialons ($\text{Si}_3\text{Al}_3\text{O}_3\text{N}_5$):

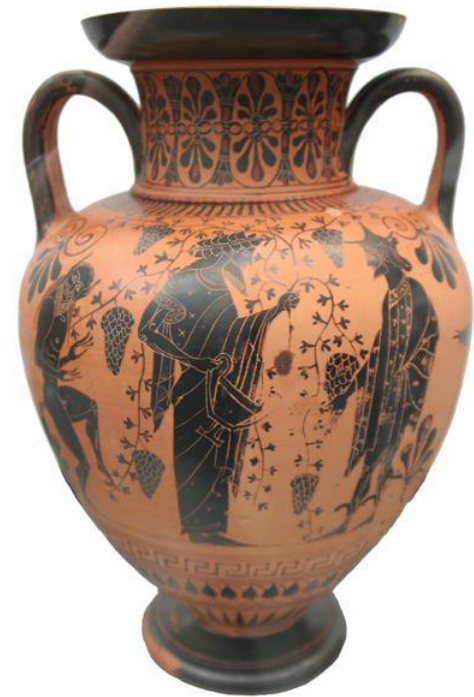
very good resistance to thermal shock
can be very tough, cutting tools

Glasses (fused silica, optical glasses, lead glasses, borosilicate, window glasses):

transparent, cheap
Low "melting" temperature, easy moulding
can be treated to improve mechanical properties

Ceramics

Traditional Ceramics



Clayware, pottery,
porcelain, mosaic, glass,
stone, bricks and tiles,
sand, earth.....

Ceramics

Engineering Ceramics

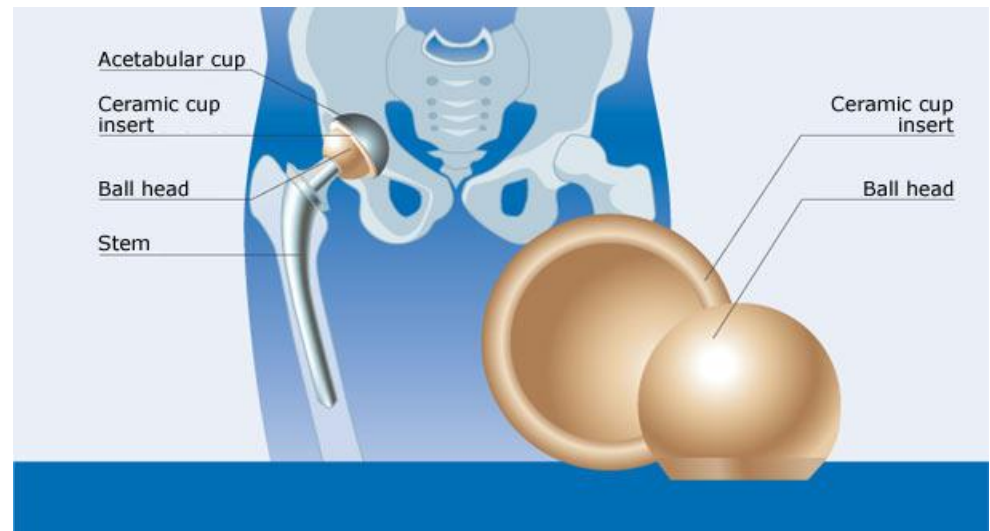
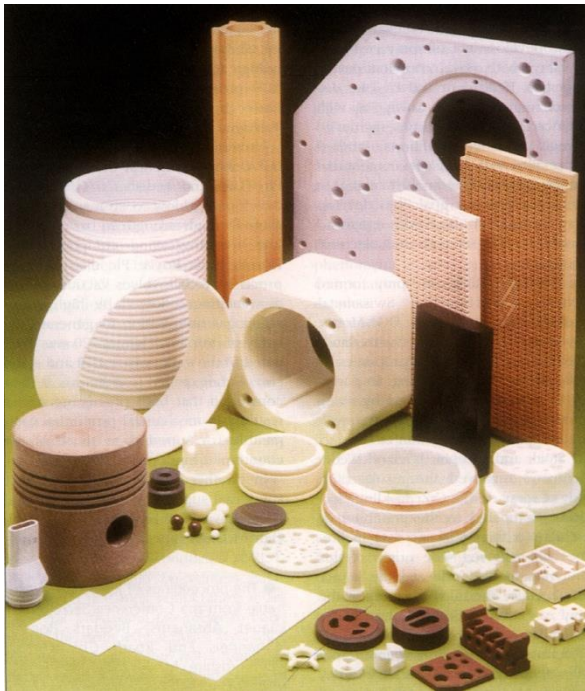
Cement, concrete,
refractories, engineering
glasses, ceramic tiles,
ceramic coatings



Ceramics

Advanced Ceramics

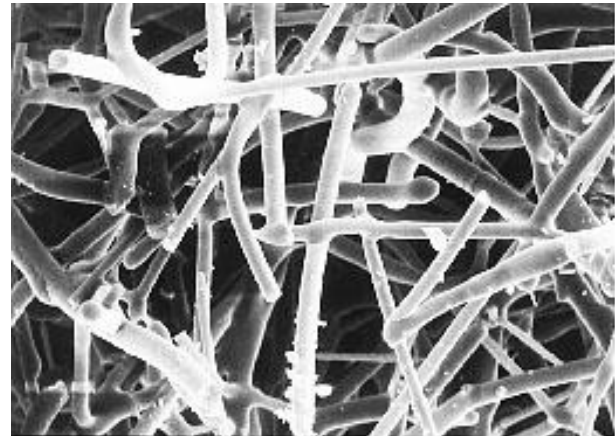
cutting tools, bioceramics, components in capacitors, resistors, insulators, piezoceramics, magnets, superconductors, catalysts, electrolytes, semiconductors, thermal protection systems, armours



Ceramics

Advanced Ceramics

thermal protection systems, armours



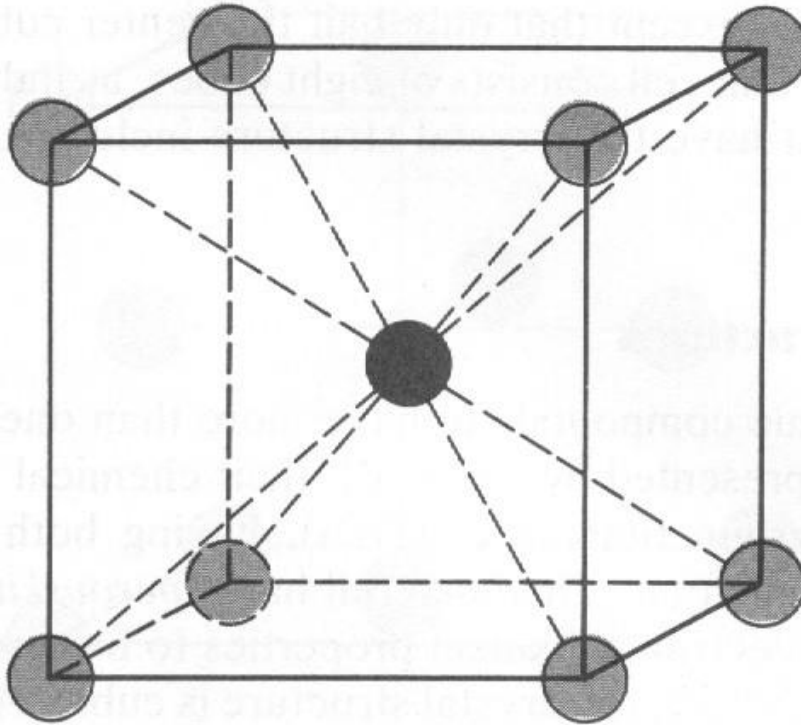
Ceramics

Crystalline Ceramics

Structure	Anion Packing	Examples
AX	FCC	NaCl, MgO, FeO
AX	Simple Cubic	CsCl
AX	FCC	ZnS, SiC
AX ₂	Simple Cubic	CaF ₂ , UO ₂ , ThO ₂
ABX ₃	FCC	BaTiO ₃ , SrZrO ₃
AB ₂ X ₄	FCC	MgAl ₂ O ₄ , FeAl ₂ O ₄

Ceramics

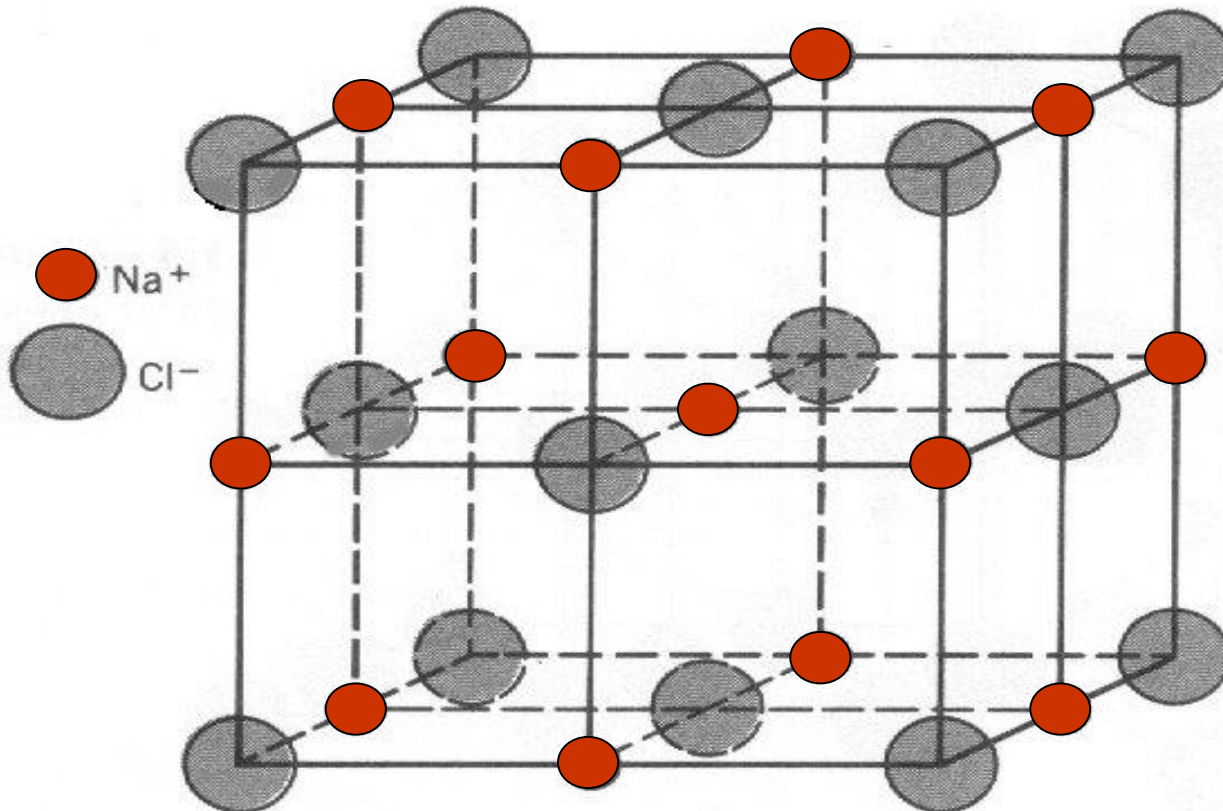
AX Structure - CsCl



Simple Cubic Crystal

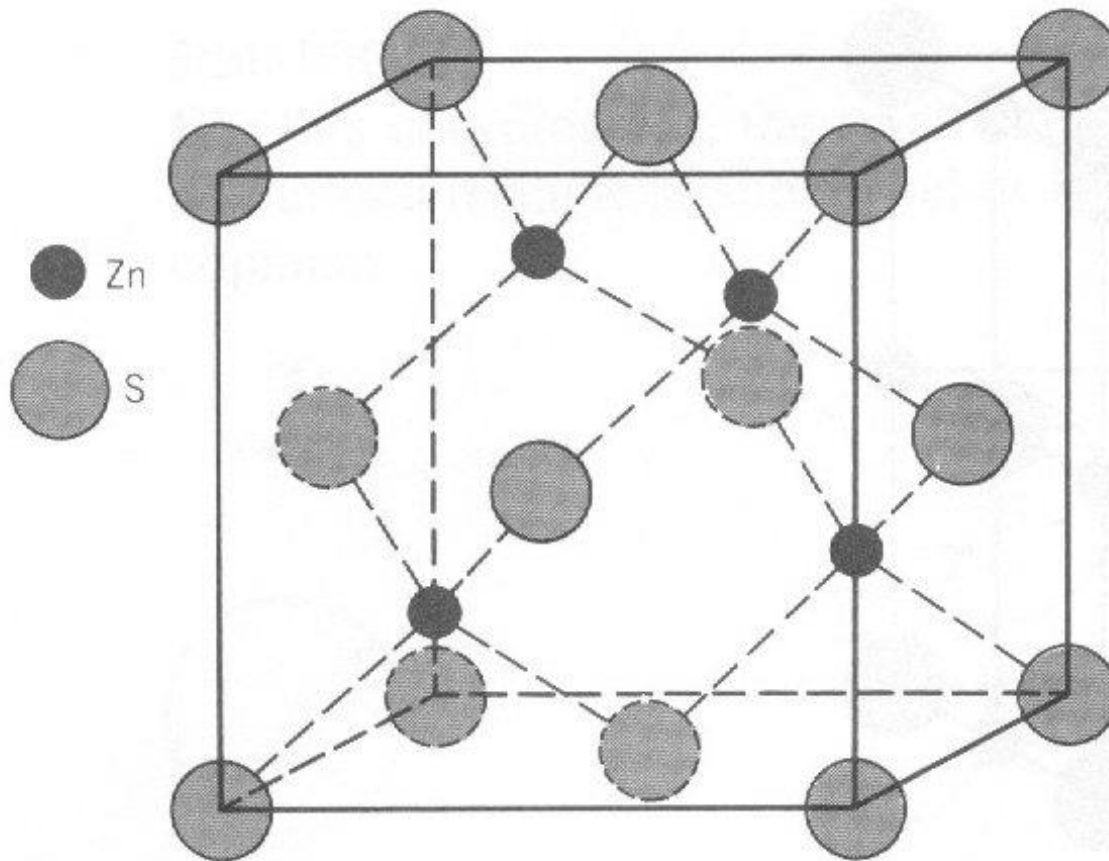
Ceramics

AX Structure - NaCl



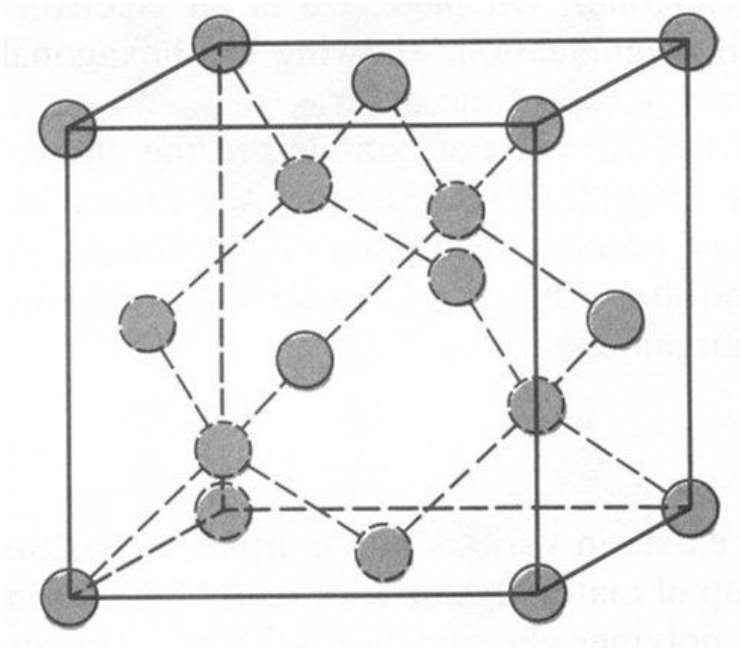
Ceramics

AX structure - Si C



Ceramics

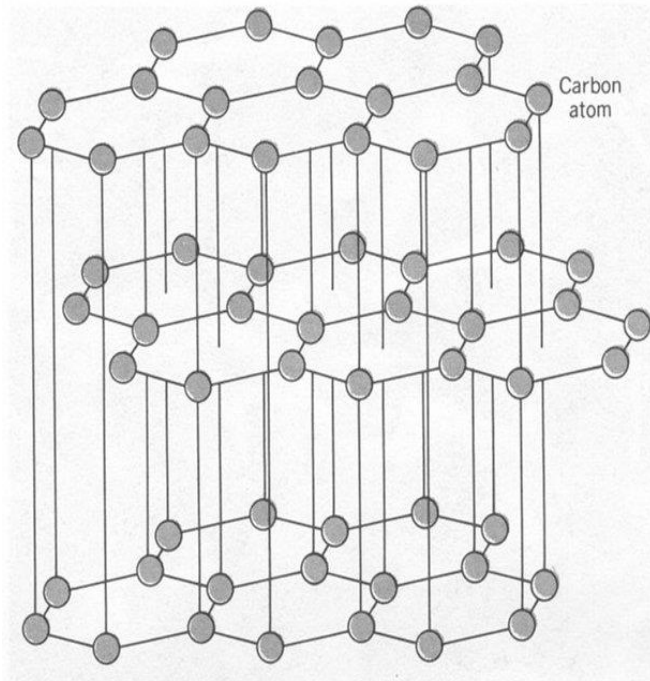
Diamond



- AX type crystal structure similar to that of ZnS.
- Each carbon atom is covalently bonded to four other C atoms in a *diamond-cubic* crystal structure.

Ceramics

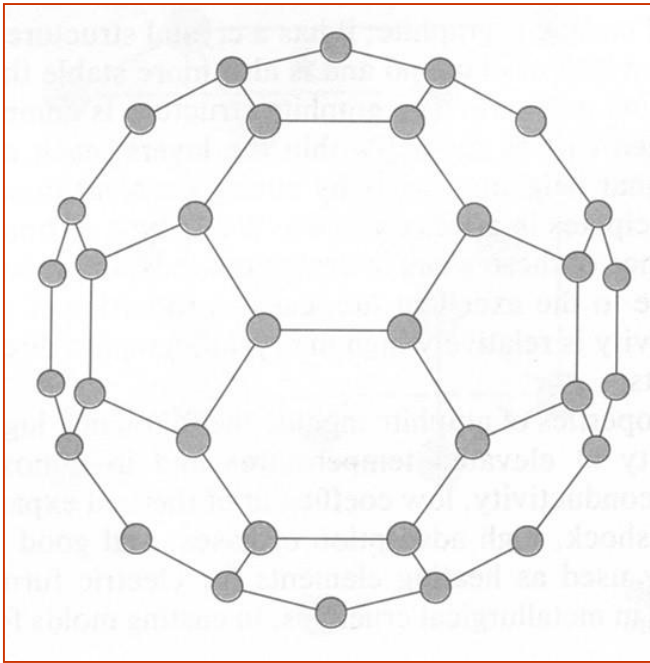
Graphite



Layers of hexagonally arranged and covalently bonded C atoms.
Between layers, weaker Van der Waals bonds are active, giving easy slip on the {0001} crystallographic planes.

Ceramics

Fullerenes - C₆₀



- Molecular form of carbon with a hollow spherical structure resembling a geodesic dome (soccer ball.)
- Called *buckyballs* after R. Buckminster Fuller, who pioneered the geodesic dome.
Discovered in 1985 and have since been found to occur naturally in several sources.
- In the solid crystalline state, C₆₀ molecules pack together in a FCC unit cell arrangement with a lattice parameter $a=1.41$ nm.